



Edge Appliance VM

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This content applies to release 2025 of Cumulocity.

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INTRODUCTION

Learn about the conceptual aspects of Cumulocity Edge, the single-server variant of the Cumulocity platform.

OVERVIEW

Edge is delivered as a software appliance designed to run on industrial PC's or local servers.

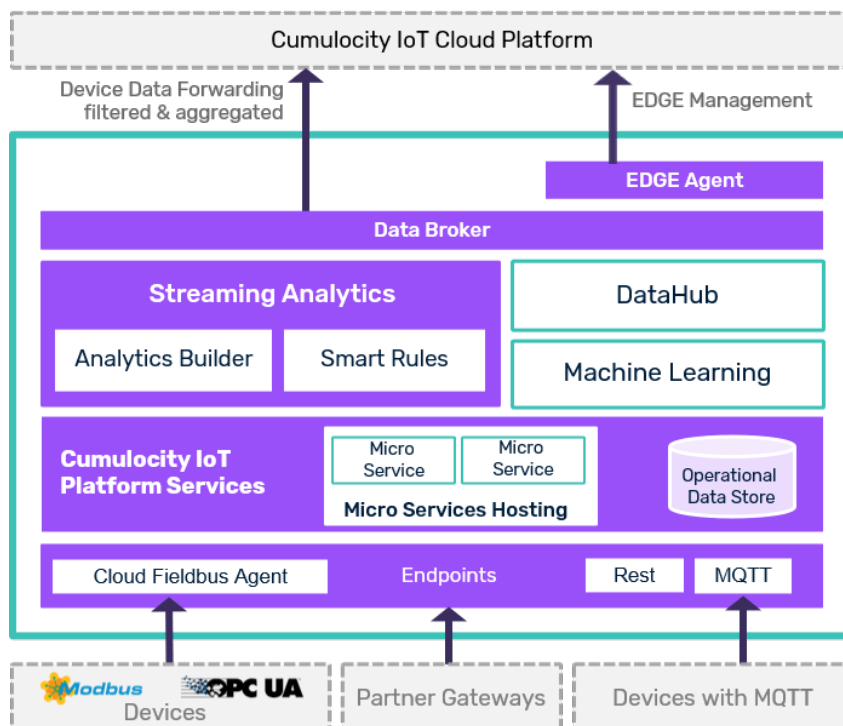
In contrast to Cumulocity platform, which is available in the cloud (for example, using AWS, Azure or other data centers), Edge is installed in factories, that is, in the same site ("onsite") in which the IoT assets are located.

Reasons for using an onsite installation of Edge include:

- **Autonomy:** Even if there is no cloud connection, tasks like data collection and data analysis can still be performed.
- **Data reduction:** Data is analyzed and aggregated close to assets, and thus less data needs to be sent to the cloud.
- **Reactivity:** Both Edge and Cumulocity platform include real-time streaming analytics engines. However, placing the rule execution in Edge reduces latency, because the round-trip to cloud is omitted.

Features of Edge include:

- Edge Agent, which enables remote monitoring and management of an Edge instance from the Cumulocity tenant.
- Data Broker to send IoT data to the cloud and receive operations from the cloud, with web-based UI to filter data.
- Streaming Analytics engine for real-time local data analysis including the Cumulocity Analytics Builder.
- Ready-to-use Cockpit and Device Management applications.
- Native protocol support for MQTT and REST.
- Edge database for operational data storage.
- **OPC UA** with web-based UI for efficient connection management and seamless integration of industrial automation systems.
- Easy installation, upgrades and backup/restore.
- Microservice hosting, which allows to run server-side applications which may be used to extend the Cumulocity platform with customer-specific functionality (optional component).
- **DataHub Edge** for historical data storage and analytics (optional component).



VERSIONS

Cumulocity Edge 10.18 uses the following versions:

Edge component	Version
Cumulocity platform	10.18.0
Streaming Analytics	10.18

CUMULOCITY EDGE VERSUS OTHER CUMULOCITY DEPLOYMENTS

Edge uses the same software as Cumulocity platform.

However, while the base software is the same, there are differences regarding the activated optional features and pre-installed agents.

The following differences apply:

Area	Edge	Cumulocity platform
Multi-tenancy	No; single tenant	Yes
Cluster	No; single server	Yes
High availability	HA capabilities depend on the underlying virtualization technology, server failure could lead to temporary downtime*	Full HA: No downtime on server failure, optionally even for data center failure
Vertical scalability	Yes, limited to appr. 100 tps per CPU core	Yes, but not used
Horizontal scalability	No	Yes, nearly unlimited scalability
Upgrades with no downtime	No	Yes
Root access	Yes	Yes, if customer is hosting
Installation	Offline, with VM image	Online
Cloud Field Bus	Included	Optional
Streaming Analytics	Included	Optional
Data Broker	Included	Optional
OPC UA	Included	Optional
Microservice Hosting	Optional	Optional

Area	Edge	Cumulocity platform
Machine Learning	Optional	Optional
Data Hub	Optional	Optional

Footnote: * The underlying infrastructure including the virtualization is not part of the product. For further details on high availability or fault tolerance options, refer to the relevant information provided by third parties (for example, [vSphere Availability](#)).

In addition to Edge, we provide a light-weight approach to implement Edge functionality: the open-source Cumulocity Device SDKs. These can be utilized to build agents on less powerful devices like embedded devices.

EDGE FUNCTIONALITIES

Since Edge is based on the same software as the cloud-based Cumulocity platform version, the included applications are the same in both versions, with minor restrictions.

For general information about functionalities and features of Cumulocity platform, see [Get familiar with the UI](#).

Below you can find the links to the most common functionalities of Cumulocity.

DEVICE MANAGEMENT

The [Device Management application](#) provides functionalities for managing and monitoring devices and enables you to control and troubleshoot devices remotely.

The following functionalities are available in Edge and described in detail in these sections:

SECTION	CONTENT
Registering devices	How to register one or more devices manually and how to bulk-register devices in order to connect devices to your account.
Viewing all devices	What is displayed in the device list and how to sort devices by filtering devices .
Grouping devices	Why and how to group devices into top-level groups, subgroups and smart groups .
Viewing device details	Detailed description of the various kind of information available for various types of devices.
Monitoring and controlling devices	How to monitor the connection quality and availability status of devices, how to handle alarms from devices, how to remote control and how to troubleshoot devices.
Managing device services	How to monitor the data that your devices send about the services they are running. See detailed information about measurements , events and alarms sent for every service.
Managing device types	How to process data from various device types by using device protocols.

SECTION	CONTENT
Managing device data	How to retrieve and manage firmware and software for devices; how to retrieve configuration data and store and manage it in a configuration repository as configuration snapshots.
SmartREST T templates	How to work with SmartREST templates , a collection of request and response templates used to convert CSV data and Cumulocity Rest API calls.
Working with simulators	Create a simulator with the same level of functionality as connected hardware devices and simulate messages (measurements, alarms, events and inventory) that the device sends to the Cumulocity platform with series of instructions that the simulator executes one after the other.
Connectivity	How to set up your Jasper Control Center account , configure the connectivity for the SIM provider in your Cumulocity tenant, link SIMs and mobile devices , and checking connectivity .

COCKPIT

The [Cockpit application](#) provides you with options to manage and monitor Internet of Things (IoT) assets and data from a business perspective.

The following functionalities are available in Edge and described in detail in these sections:

Section	Content
Managing assets	Organize assets in hierarchies by creating groups and assigning devices .
Data Explorer	Interactively explore, compare and visualize IoT data. Describes how to access and use the data explorer , add data points to the data explorer, customize data point properties , modify the visualization , store the data explorer as widget , and export the data.
Working with dashboards	Create your own analytics and monitor pages by adding and arranging widgets . Share dashboards among all devices of the same type.
Working with reports	Handle reports based on dashboard layouts, create reports for exporting data in CSV or excel format and schedule the export .
Using widgets in dashboards and reports	Widgets can display maps, images, graphs, tables and other graphic representations of data. Add a widget to a dashboard or a report , modify widgets, and add assets to a new or existing widget .
Widgets collection	Use various types of widgets from the Widgets collection that comes with Edge and configure them according your needs.
Managing exports	Export data to either CSV or Excel files. Add and schedule an export to a CSV or XLSX file to any point in time.
Data point library	Manage default settings ("profiles") of your devices and apply them automatically using the Data point library .
Smart Rules	Create and manage business rules to work on incoming data in realtime and to perform actions based on this data.
Smart Rules collection	Use pre-defined global Smart Rules to configure rules for geofencing, thresholds or alarm escalation and notifications (SMS/email). Describes each smart rule and its configurable parameters in detail.
Configuring Cockpit applications	Configure custom Cockpit applications according to your individual needs.

ADMINISTRATION

The [Administration application](#) enables account administrators to manage their users, roles, tenants and applications and lets them configure a number of settings for their account.

The following functionalities are available in Edge and described in detail in these sections:

SECTION	CONTENT
Configuring Edge	Providing information on Edge specific configuration options.
Home screen	Providing information on your capacity usage and subscribed applications.
Managing users	How to add users , edit, disable or delete them.
Managing permissions	How to create and edit global roles and inventory roles , how to assign them to users, and how to grant application access .
Managing applications	How to manage and configure own applications in your Edge account.
Audit logs	How to view and filter audit logs that show security-relevant operations that have been processed by a user. See the different audit log types for more information.
Alarm mapping	Change the severity and text of alarms to adapt them to your business priorities.
Managing data	How to manage and configure retention rules for your data and how to manage stored files in the file repository.
Changing settings	How to change account settings like application settings , manage the properties library , enable SMS provider features like user notifications and two-factor authentication, and manage credentials for different providers .

CONFIGURING THE EDGE INFRASTRUCTURE

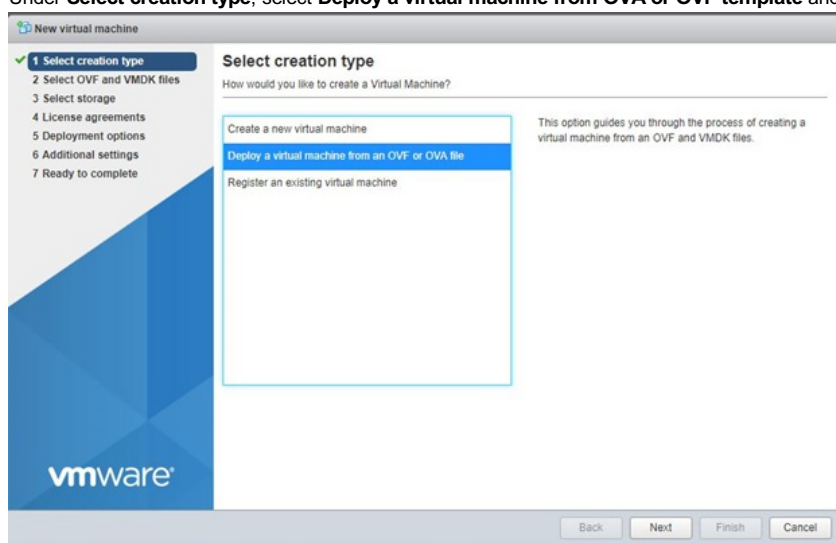
Find out how to set up the hypervisor infrastructure for Cumulocity Edge.

EXAMPLE SETUP FOR VMWARE ESXI

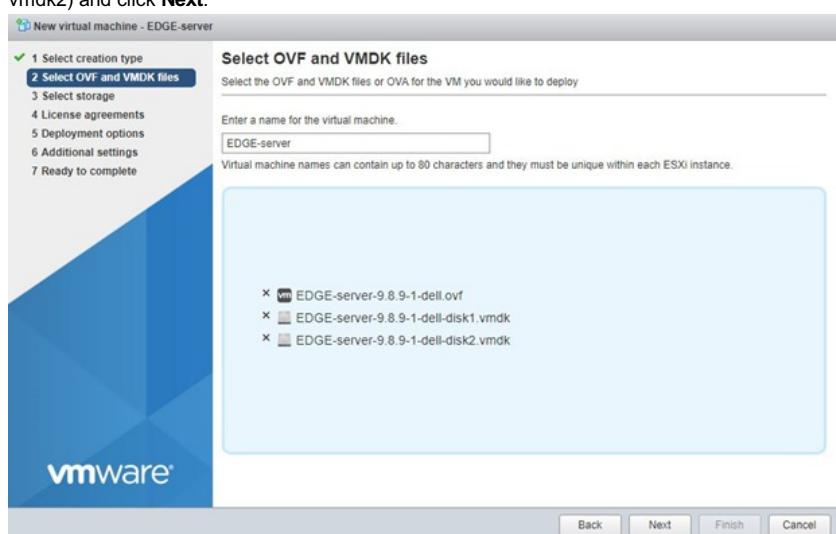
SETTING UP ESXI

To set up an ESXi virtual machine, follow these steps:

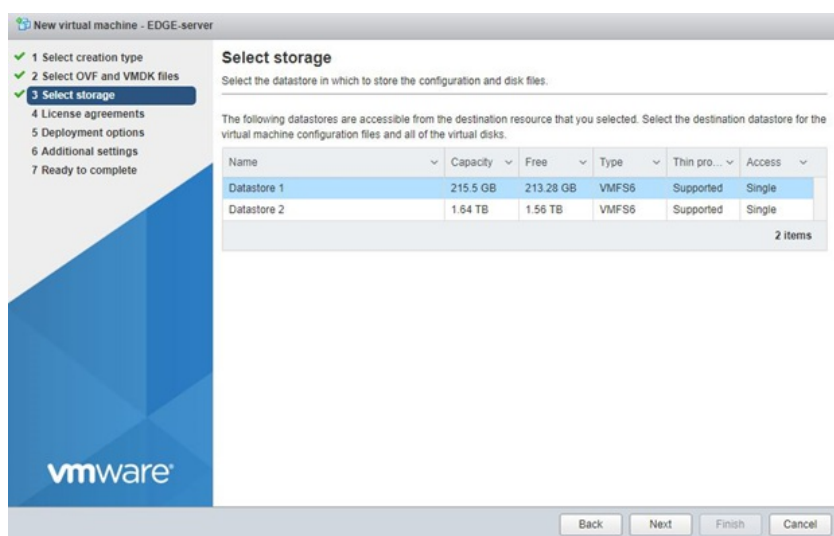
1. Click **Create/Register VM** to open a window to select the creation type.
2. Under **Select creation type**, select **Deploy a virtual machine from OVA or OVF template** and click **Next**.



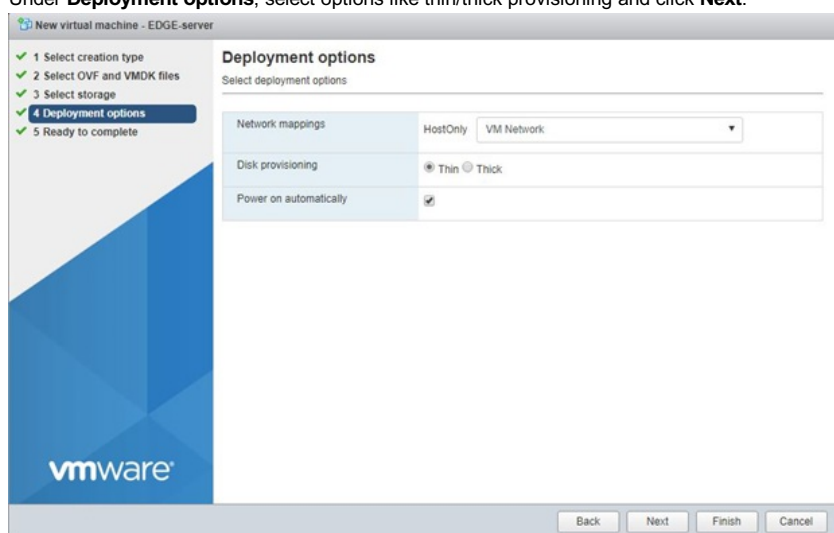
3. In the next window, provide a name for the VM, for example, EDGE-server, and drag and drop the required files (ovf, vmdk1, vmdk2) and click **Next**.



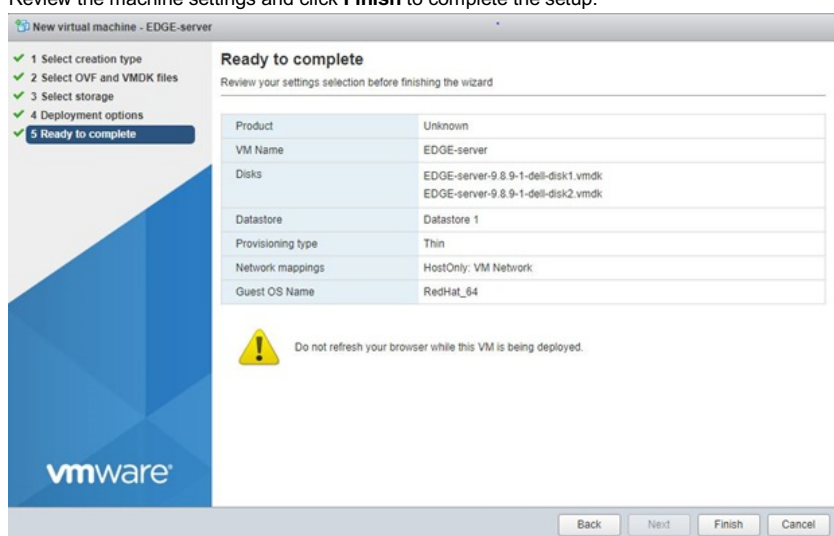
4. Under **Select storage**, select the datastore where the VM will reside and click **Next**.



5. Under **Deployment options**, select options like thin/thick provisioning and click **Next**.



6. Review the machine settings and click **Finish** to complete the setup.



A VM with the provided name, for example, "EDGE-server", should now show up in the **Virtual Machines** section. Notifications will appear accordingly in the **Recent tasks** pane.

VM HARDWARE CONFIGURATION

- Change VM compatibility to the latest ESXi version.
- Make sure that only one network interface is configured. Use VMXNET3 as type.
- Edit VM Settings -> VMware Tools: Check time synchronization
- Edit VM Settings -> General Options: Set guest OS to CentOS 7 (64bit)

Next, perform the Edge appliance installation. See, [Installing Edge](#).

EXAMPLE SETUP FOR VMWARE WORKSTATION PLAYER

SETTING UP FOR VMWARE

To set up the Edge appliance in VMware Workstation Player, follow the steps below.

INFO

The following steps show a reference example. Refer to the VMware documentation for the exact setup. The final configuration also depends on the end user setup.

1. In VMware, navigate to **Player > File > Open** to import the Edge appliance.
2. Navigate to the folder where the Edge appliance files are located, select the OVF file and click **Open**.
3. Change the Edge appliance name if required and click **Import**. You can also change the storage path of the Edge appliance here.

IMPORTANT

On VMware Workstation, you must use UTC on your host machine. If you choose not to use UTC, you may have time sync issues. Set `rtc.diffFromUTC=0` in the .vmx file to avoid the time sync issues.

4. Start the Edge appliance by clicking **Play virtual machine**.

Next, perform the Edge appliance installation. See, [Installing Edge](#).

SETTING UP FOR VMNETCFG UTILITY

You can use the VMware `vmnetcfg` utility to get the necessary details like the subnet mask and gateway IP required to configure the network.

The following example illustrates the network configuration on a Windows platform. For instructions on Linux platform, see VMware Knowledge Base.

1. Download the correct version of the `vmnetcfg` utility. It can also be extracted from the VMware Workstation Pro installer.
2. Save the vmnetcfg binary file (`vmnetcfg.exe`) in the VMware Workstation Player installation directory. In a Windows environment, this is usually `C:\Program Files (x86)\VMware\VMware Player`.
3. Open the file with the appropriate rights.

Virtual Network Editor

Name	Type	External Connection	Host Connection	DHCP	Subnet Address
VMnet0	Bridged	Auto-bridging	-	-	-
VMnet1	Host-only	-	Connected	Enabled	192.168.137.0
VMnet8	NAT	NAT	Connected	Enabled	192.168.117.0

< >

Add Network... Remove Network Rename Network...

VMnet Information

☐ Bridged (connect VMs directly to the external network)
 Bridged to: Automatic Automatic Settings...

☒ NAT (shared host's IP address with VMs) NAT Settings...

☐ Host-only (connect VMs internally in a private network)

☒ Connect a host virtual adapter to this network
 Host virtual adapter name:

☒ Use local DHCP service to distribute IP address to VMs DHCP Settings...

Subnet IP: 192 . 168 . 117 . 0 Subnet mask: 255 . 255 . 255 . 0

Restore Defaults OK Cancel Apply Help

4. Select "NAT" as external connection.
5. Click **NAT settings** to open the **NAT Settings** window.
6. Note down the gateway IP address and close the **NAT settings** window.

NAT Settings

Network: vmnet8
 Subnet IP: 192.168.117.0
 Subnet mask: 255.255.255.0
 Gateway IP: 192.168.117.2

Port Forwarding

Host Port	Type	Virtual Machine IP Address	Description
< >			

Add... Remove Properties

Advanced

☒ Allow active FTP
☒ Allow any Organizationally Unique Identifier

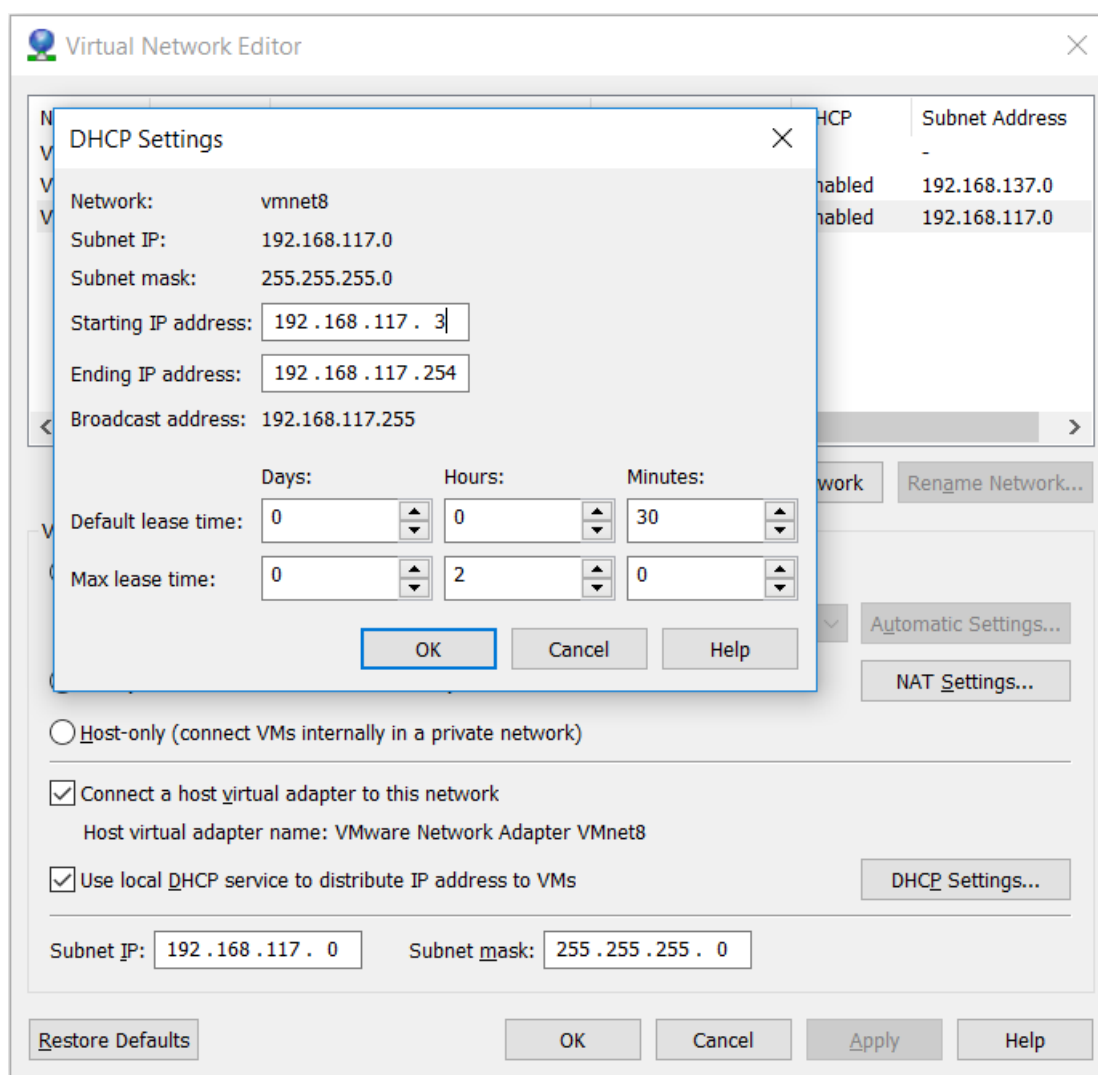
UDP timeout (in seconds): 30
 Config port: 0

☐ Enable IPv6
 IPv6 prefix: fd15:4ba5:5a2b:1008::/64

DNS Settings... NetBIOS Settings...

OK Cancel Help

7. Click **DHCP Settings** to open the **DHCP Settings** window.
8. In the fields **Starting IP address** and **Ending IP address**, change the IP range from 3 to 254, that means, if your gateway IP is 192.168.117.2, set the IP range from 192.168.117.3 to 192.168.117.254.



9. Click **OK** to save your settings.

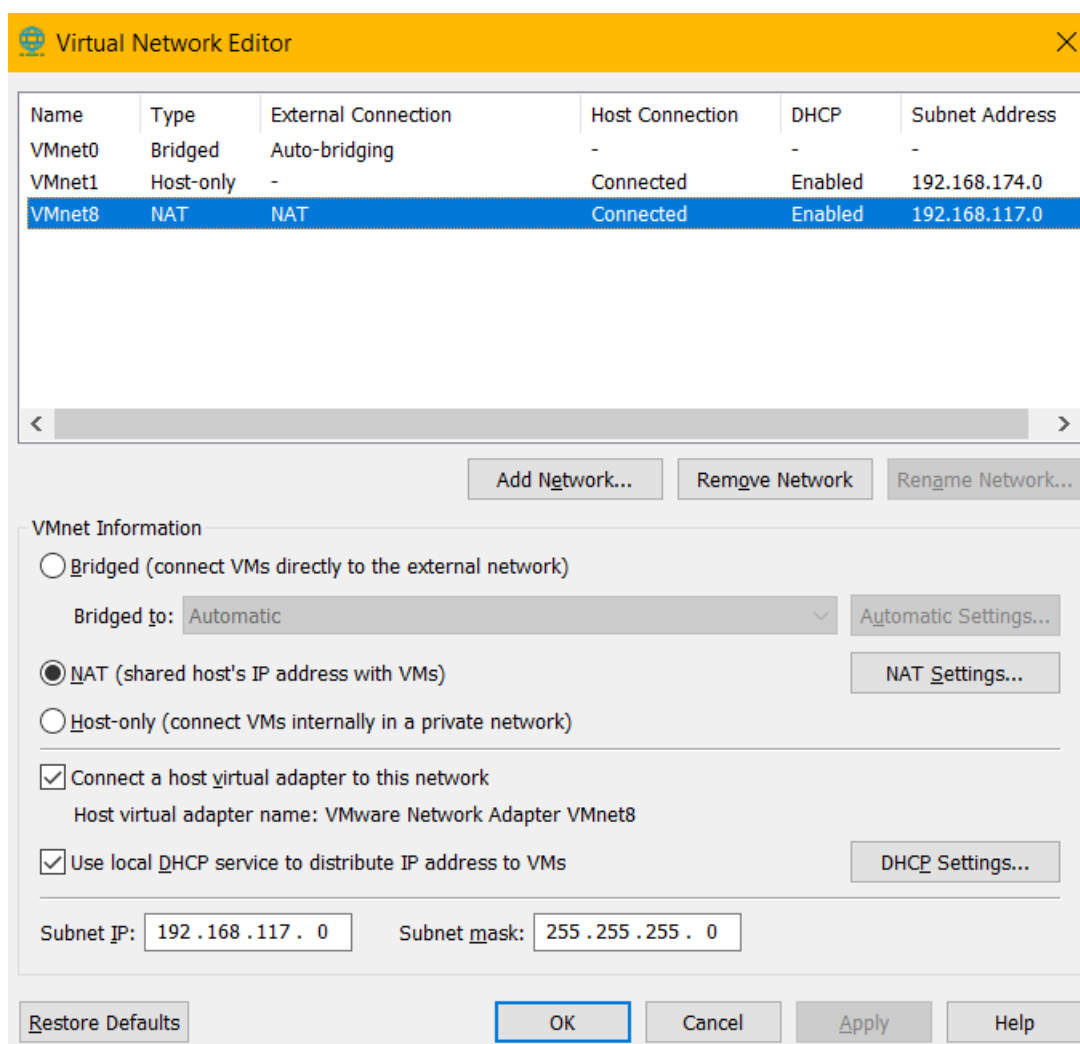
PORT FORWARDING ON A VMWARE PLATFORM

To perform port forwarding on a VMware platform, follow the steps below.

❗ IMPORTANT

The IP addresses used in the screenshots are sample IP addresses, for the purpose of example only. The IP addresses may vary in your set up.

1. Run the *vmnetcfg.exe* utility as an administrator.



2. Select the **NAT** adapter and click **NAT Settings**.

NAT Settings ✕

Network: vmnet8
 Subnet IP: 192.168.117.0
 Subnet mask: 255.255.255.0
 Gateway IP: 192.168.117.2

Port Forwarding

Host Port	Type	Virtual Machine IP Address	Description
443	TCP	192.168.117.10:443	HTTPS
1883	TCP	192.168.117.10:1883	MQTT
8883	TCP	192.168.117.10:8883	MQTT over SSL

Add... Remove Properties

Advanced

☒ Allow active FTP
☒ Allow any Organizational Unique Identifier

UDP timeout (in seconds): 30

Config port: 0

☐ Enable IPv6
 IPv6 prefix: fd15:4ba5:5a2b:1008::/64

DNS Settings... NetBIOS Settings...

OK Cancel Help

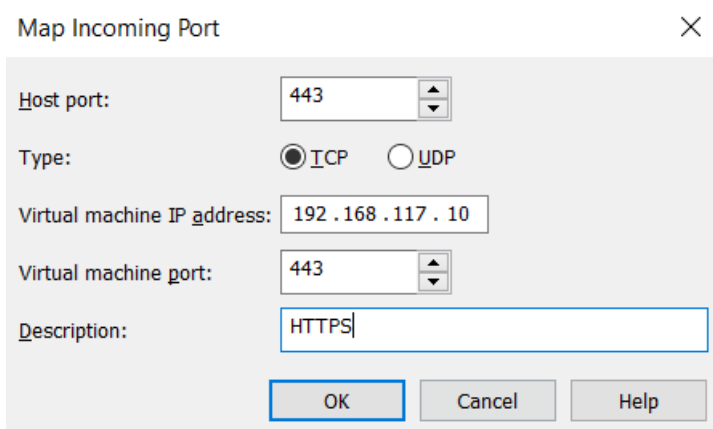
3. Click **Add** to add the port forwarding rules. Enter the port forwarding rules for HTTPS, MQTT and MQTT over SSL protocols as described in the table below:

i INFO

Ensure that the **Virtual machine IP address** is the same as the IP address that you have configured for the Edge appliance.

Host Port	Type	VM IP Address	VM Port	Description
443	TCP	192.168.117.10	443	HTTPS
1883	TCP	192.168.117.10	1883	MQTT
8883	TCP	192.168.117.10	8883	MQTT over SSL

Click **OK** in the **Map Incoming Port** window after entering the port forwarding rules for each port.



The dialog box titled "Map Incoming Port" contains the following fields and controls:

- Host port:** A numeric input field with the value 443.
- Type:** Two radio buttons, with **ICP** selected and **UDP** unselected.
- Virtual machine IP address:** A text input field containing the IP address 192.168.117.10.
- Virtual machine port:** A numeric input field with the value 443.
- Description:** A text input field containing the text HTTPS.
- Buttons:** Three buttons at the bottom: **OK**, **Cancel**, and **Help**.

4. Click **OK** in the **NAT Settings** window.
5. Click **Apply > OK**.

EXAMPLE SETUP FOR HYPER-V

To set up Hyper-V, you must first enable Hyper-V on your system, and create Network Address Translation (NAT) adapter. For enabling Hyper-V, see the Microsoft documentation.

CREATING A NAT ADAPTER

You can create a NAT adapter using two ways:

- [Using default settings](#)
- [Using user-defined settings](#)

i INFO

Run all the commands in Windows PowerShell with administrative privileges.

CREATING A NAT ADAPTER USING DEFAULT SETTINGS

i INFO

You must use the default adapter name **NATSwitchForEdge** and the default IP address **192.168.66.1**.

1. Open Windows PowerShell with administrative privileges.
2. Create a new NAT adapter.

```
PS C:\WINDOWS\system32> New-VMSwitch -SwitchName "NATSwitchForEdge" -SwitchType Internal
```

3. Verify that the new NAT adapter is created.

```
PS C:\WINDOWS\system32> Get-NetAdapter
```

```
PS C:\WINDOWS\system32> Get-NetAdapter
```

Name	InterfaceDescription	ifIndex	Status	MacAddress	LinkSpeed
Ethernet (NATSwitchFo...	Hyper-V Virtual Ethernet Adapter #2	31	Up	00-15-50-38-01-49	10 Gbps
Ethernet	Intel(R) Ethernet Connection I217-LM	26	Up	20-47-47-B2-A0-6C	1 Gbps
VMware Network Adapte...	VMware Virtual Ethernet Adapter for ...	23	Up	00-50-56-C0-00-08	100 Mbps
Ethernet (Default Swi...	Hyper-V Virtual Ethernet Adapter	30	Up	2E-15-FF-77-7F-44	10 Gbps
Wi-Fi	Intel(R) Centrino(R) Advanced-N 6235	19	Disconnected	C4-D9-87-F9-AA-88	6 Mbps
Bluetooth Network Conn...	Bluetooth Device (Personal Area Netw...	16	Disconnected	C4-D9-87-F9-AA-8C	3 Mbps
VMware Network Adapte...	VMware Virtual Ethernet Adapter for ...	6	Up	00-50-56-C0-00-01	100 Mbps
Barracuda VPN	phion Virtual Adapter (VPN)	3	Not Present	50-48-49-4F-4E-43	0 bps
VirtualBox Host-Only ...	VirtualBox Host-Only Ethernet Adap...#3	2	Not Present	0A-00-27-00-00-02	0 bps

Note down the `ifIndex` of the adapter. The `ifIndex` in the above screenshot is 31. The `ifIndex` may vary in your system and the same `ifIndex` should be used in the next step.

- Assign an IP address for the adapter.

```
PS C:\WINDOWS\system32> New-NetIPAddress -IPAddress 192.168.66.1 -PrefixLength 24 -InterfaceIndex 31
```

- Create a NAT rule.

```
PS C:\WINDOWS\system32> New-NetNat -Name NATSwitchForEDGE -InternalIPInterfaceAddressPrefix 192.168.66.0/24
```

CREATING A NAT ADAPTER USING USER-DEFINED SETTINGS

You can define the adapter name and configure the IP address for your Edge appliance. In this example, the adapter name is `EdgeAdapter1` and the IP address is `10.20.30.40`.

INFO

The following steps show a reference example. The final configuration also depends on the end user setup.

- Open Windows PowerShell with administrative privileges.
- Create a new NAT adapter.

```
PS C:\WINDOWS\system32> New-VMSwitch -SwitchName "EdgeAdapter1" -SwitchType Internal
```

- Verify that the new NAT adapter is created.

```
PS C:\WINDOWS\system32> Get-NetAdapter
```

Note down the `ifIndex` of the adapter. The `ifIndex` may vary in your system and the same `ifIndex` should be used in the next step.

- Assign an IP address for the adapter.

```
PS C:\WINDOWS\system32> New-NetIPAddress -IPAddress 10.20.30.40 -PrefixLength 24 -InterfaceIndex 71
```

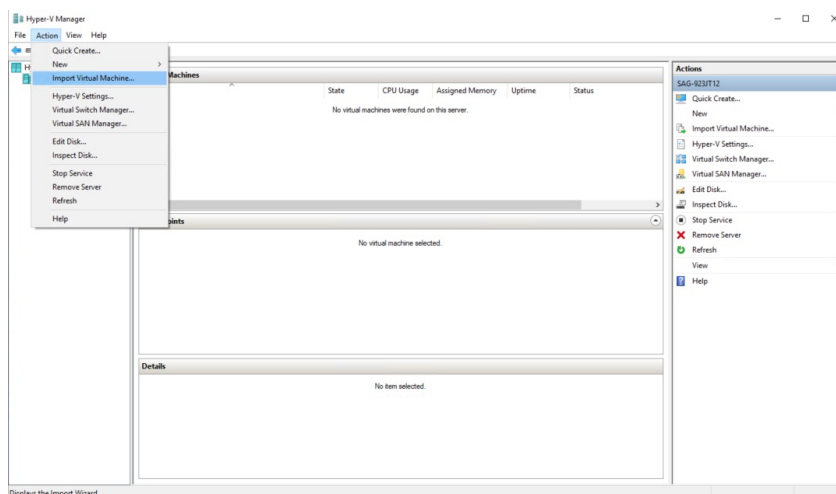
- Create a NAT rule.

```
PS C:\WINDOWS\system32> New-NetNat -Name EdgeNATRule1 -InternalIPInterfaceAddressPrefix 10.20.30.0/24
```

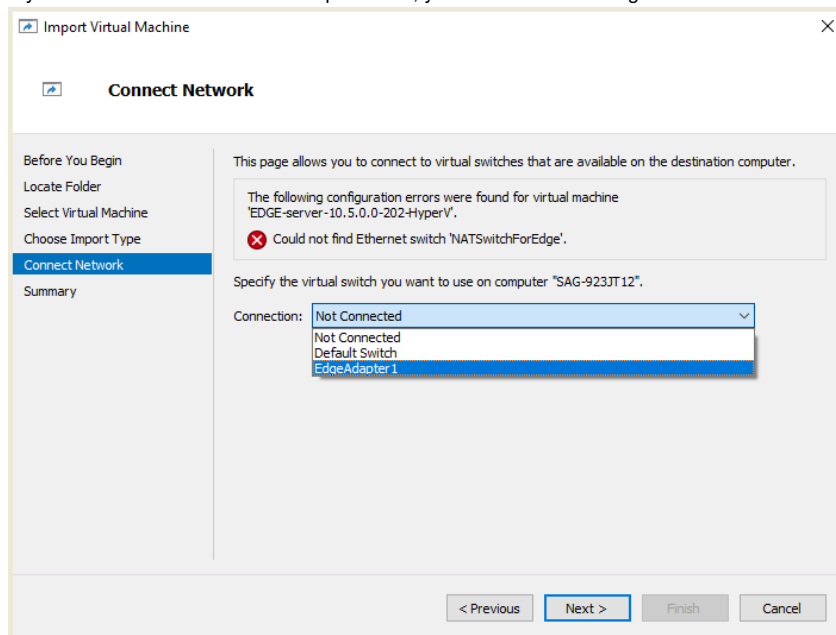
SETTING UP THE EDGE APPLIANCE IN HYPER-V

After creating a NAT adapter, you must import the Edge appliance image into Hyper-V.

- In Hyper-V Manager, go to **Action > Import Virtual Machine**.

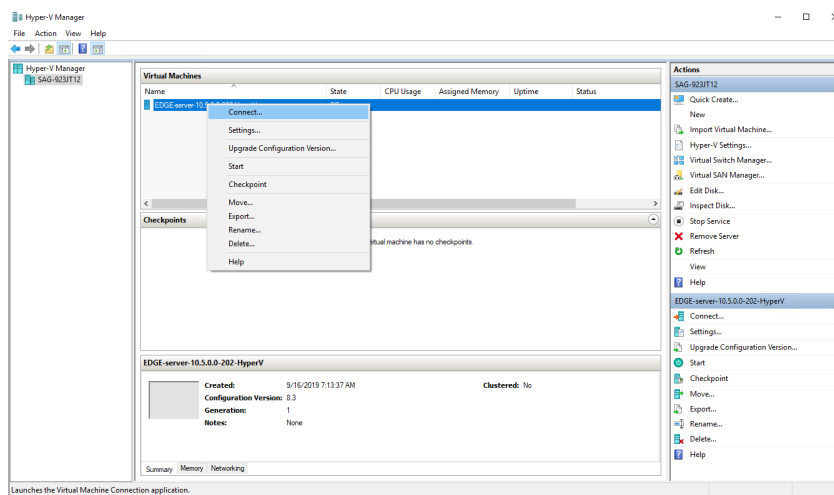


2. In the **Before You Begin** wizard, click **Next**.
 3. Provide the location of the Edge appliance image and click **Next**.
 4. Select the Edge appliance and click **Next**.
 5. In the **Choose Import Type** wizard, select **Register the virtual machine in-place (use the existing unique ID)** and click **Next**.
 6. If you have used the default NAT adapter name, skip this step and proceed with step 7.
- If you have used a different NAT adapter name, you will see the following screen:



Select the adapter from the

- Connection** drop down list and click **Next**.
7. Click **Finish**. The Edge appliance image appears in the Hyper-V Manager.
 8. Right-click the Edge appliance image and click **Connect**.



INFO

By default, the NTP servers are configured to Europe NTP servers. You can configure the NTP server as per your requirements.

INSTALLING EDGE

Follow along this section to learn about the requirements and steps for installing Cumulocity Edge.

REQUIREMENTS

INSTALLATION REQUIREMENTS

Item	Details
Hypervisor	- The VMWare image is set with hardware version 13, which is compatible with ESXi 6.5, ESXi 6.7, ESXi 6.7 Update 2, ESXi 6.7 Update 3, ESXi 7.0, ESXi 7.0 Update 1, ESXi 7.0 Update 2, ESXi 7.0 Update 3, ESXi 8.0, ESXi 8.0 Update 1, and ESXi 8.0 Update 2. - VMware Workstation Player 17.x and VMware Workstation Pro 17.x - Hyper-V on Microsoft Windows 10 Enterprise, version 1809. The supported VM configuration version is 9.0. Important: Ensure that you keep the virtualization platform updated and free from vulnerabilities by following the security advisories provided by the vendors of the hypervisor.
Edge appliance image	To be downloaded from the Cumulocity Download Center, based on the target hypervisor. For VMware (ESXi and Workstation Player), download all the 4 files: - Cumulocity Edge - VMWare [Disc 1] (<i>version</i>) - Cumulocity Edge - VMWare [Disc 2] (<i>version</i>) - Cumulocity Edge - VMWare .mf (<i>version</i>) - {Cumulocity Edge - VMWare .ovf (<i>version</i>) For Hyper-V, download the ZIP file: - Cumulocity Edge - HyperV (<i>version</i>) The fixes for Edge: - Cumulocity Edge (<i>version</i>) Update
Edge license file	To request the license file for Edge, contact product support In the email, you must include - your company name, under which the license has been bought - the domain name (for example, myedge.domain.com), where Edge will be reachable For more information, see Domain name validation for Edge license key generation.
SSL key and SSL certificate	Optional. Use your internal or an external CA (Certification Authority) to generate these files. These files must not be password-protected. Ensure that the SSL certificate has the complete certificates chain in the right order. Info: The .crt and .key files must be in the PEM format and the .key file must not be encrypted.
DNS entry	The DNS (Domain Name System) is used to resolve human readable host names like www.cumulocity.com to machine readable IP addresses like 192.198.1.10. If you want to connect to Edge appliance within your LAN, the DNS entry must be added for the domain name (URL under which Edge can be reached) with the IP address of the host.
Edge cloud remote access	To connect and manage one (or multiple) Edge appliances to your Cumulocity cloud tenant, you need an active Cumulocity Standard tenant with the Data Broker and Cloud Remote Access extensions. Info: The Edge cloud remote access is an optional feature in Edge.

DOMAIN NAME VALIDATION FOR EDGE LICENSE KEY GENERATION

To procure the Edge license file, you must provide the right domain name to the Cumulocity's logistics team for Edge license key

generation.

When you provide the domain name, consider the following points:

- The domain name does not need to be a Fully Qualified Domain Name (FQDN). For example, to access the Edge appliance with the domain name **myown.iot.com**, request the Edge license for **myown.iot.com** or **iot.com** (without the subdomain **myown**).
- If you exclude the subdomain from the domain name in the Edge license, you must possess a wildcard SSL certificate which can be used with multiple subdomains (**myown** or others) of the domain (**iot.com**).
For example, if you provide **iot.com** as the domain name, you must possess an SSL certificate for **.iot.com**.
- If you have an Internationalized Domain Name (IDN), then you must provide the translated ASCII equivalent domain name. For example, if your domain name is **myown.iöt.com** (for example, containing **ö**), then you must use **myown.xn--it-fka.com**. Also, provide the same translated ASCII equivalent domain name as the tenant domain name during the Edge installation process.

INFO

An Internationalized Domain Name (IDN) is an internet domain name that contains at least one label, in whole or in part, in a language-specific script or alphabet, such as Arabic, Chinese, Cyrillic, Devanagari, Hebrew or the Latin alphabet-based characters with diacritics or ligatures, such as French. The internationalization of domain names is a technical solution to translate names written in language-native scripts into an ASCII text representation that is compatible with the Domain Name System. See [Wikipedia](#).

- Ensure that you adhere to the following domain name validation rules:
 - The domain name must be a combination of case-insensitive alphanumeric characters separated by dot (.) or hyphen (-).
 - Cannot contain any letters of languages like Chinese, Latin or Arabic.
 - Cannot contain any special characters like (+ , ! @ # \$ % ^ & * () ; \ \ / | < > \ " ') other than dot (.) or hyphen (-).
 - The length of the domain name including the dot must not exceed 255 characters.
 - The domain name must contain at least one dot.
 - Each segment of the domain name must be separated by a dot.
 - The domain name must be between 1 to 63 characters long.
 - The Top-Level Domain (TLD) which refers to the last segment of the domain name must be between two to six characters long.
 - The domain name cannot begin or end with a hyphen.

NETWORK CONNECTIVITY

The following network ports must be reachable from the local network:

- HTTPS
- MQTT over TLS
- SSH, only for configuring the appliance

If Edge should communicate with the cloud, the following ports of [www.cumulocity.com](#) (or another instance) must be available:

- HTTPS
- MQTT over TLS

There is no internet connection required during installation. Internet connection during runtime is optional, and only used if this is configured in the Data Broker.

Incoming traffic

The following ports must be enabled by default in order to accept traffic from users and devices on the internet (see [Setting up port forwarding](#)):

Source IP	Source Port	Destination IP	Destination Port	Service
any	any	Edge appliance IP	TCP/443	HTTPS
any	any	Edge appliance IP	TCP/1883	MQTT
any	any	Edge appliance IP	TCP/8883	MQTT/TLS

Depending on additional integrations more ports must be opened.

Outgoing Traffic

The core node must be able to connect to the internet. Ports required to outside are:

Service	Port
HTTPS	443

Depending on the installed integrations (email, SMS, and so on), different ports might have to be opened in order to make these services available to the Cumulocity platform. The descriptions of these ports are delivered with the corresponding integrations.

Depending on the DNS and NTP setup it might be the case that DNS (UDP/53) and NTP (UDP/123) connections to the internet must be possible from all hosts.

HARDWARE REQUIREMENTS

The virtual machine has the following minimum hardware requirements:

Hardware	Configuration
Disk space	100 GB
CPU	Intel x86
Number of CPU cores	2 - without microservice 4 - with microservice
RAM	6 GB - without microservice 8 GB - with microservice
Network Interface Controller (NIC)	1

These are the minimum system requirements to enable the microservice hosting feature. If the microservice requires additional system resources, you must configure the system requirements accordingly in addition to minimum system requirements. For example, if the microservice requires 2 CPU cores and 4 GB of RAM, then the VM must have 6 CPU cores (4 cores for VM + 2 cores for microservice) and 12 GB of RAM (8 GB for VM + 4 GB for microservice).

INFO

This does not cover host operating system hardware requirements. The host operating system resource requirements must be sized independently and should be over and above the resource allocated to the virtual machines.

INSTALLING EDGE USING THE UI

To install Edge using the user interface:

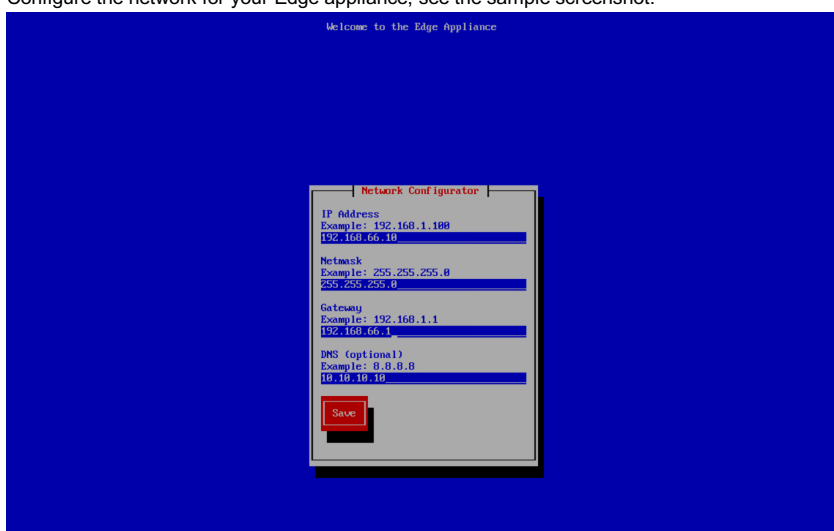
1. Connect and start the Edge appliance in the hypervisor. Wait until the network configuration screen appears.



i INFO

The network configuration blue screen does not appear post installation. After the installation, you must use the Administration application or the REST API to configure the network.

2. Configure the network for your Edge appliance, see the sample screenshot.



i IMPORTANT

While configuring the network on VMware based hypervisors, do not use the IP addresses:

- X.X.X.1 (used by VMnet8 network adapter on host)
- X.X.X.2 (used by VMnet8 network adapter's gateway)
- X.X.X.254 (used by VMnet8 network adapter's DHCP server)

3. Press **Enter** to save the network configuration.



Note down the URL to perform the installation. In the screenshot above, the URL is <https://192.168.66.10/apps/installation/>.

4. Open the URL in a browser to start the installation process.

Read the prerequisites and ensure that you have the domain name, SSL certificate and key associated with your domain name, and the license file.

5. Click **Start Installation**.
6. Create an administrator account for the guest operating system below **Guest OS admin**.
7. Provide a password for the root user of the guest operating system below **Guest OS root**, and click **Next**.

❗ IMPORTANT

Do not use the root credentials to perform any task. The root credentials must be used only when asked by Cumulocity support personnel. Using it otherwise might void the appliance support.

8. Create an administrator account to access the "edge" tenant and the Management tenant, and click **Next**.
9. Provide a fully qualified domain name below **Domain name**.

For example, "myown.iot.com". Here, you must have the Edge license for the domain name **iot.com** or **myown.iot.com**.

The domain name must adhere to all the domain name validation rules as described in [Domain name validation](#).

10. Provide the Edge license file associated with your domain name below **Licence file**.
11. Provide the SSL certificate file and the SSL certificate key file.

If you do not have an SSL certificate, select **Generate self-signed certificate** to generate one.

12. Click **Install**.

During the installation, the certificates are updated in the Edge appliance. If these certificates are not accepted by your browser, the browser does not get the progress of the installation. In such case, you must refresh the browser and follow the browser instructions for more details. The installation takes some time to complete. After the installation is complete, the "Cumulocity Edge installation is now complete" message appears.

❗ INFO

If you still see the installation in progress, refresh the browser.

Next, click **Open Cumulocity Edge**.

❗ IMPORTANT

In case you need to reset the password, you must configure the “reset password” template and email server settings to receive the password reset email. For more information, see [Password reset](#) and [Email server](#).

INSTALLING EDGE USING THE REST APIS

To install Edge using the REST APIs, see the [edge/install](#) API in the [Cumulocity Edge OpenAPI Specification](#).

ACCESSING THE EDGE APPLIANCE

You access the Edge appliance using a [domain name](#) in a web browser.

SUPPORTED BROWSERS

Supported browsers in this version are:

- Microsoft Edge (latest Chromium-based version)
- Mozilla Firefox (latest Extended Support Release [1])
- Google Chrome [2]

[1] Only the latest Extended Support Release of Mozilla Firefox is explicitly supported. Possible incompatibilities will be removed during the regular maintenance process of Cumulocity. Due to frequent upgrades of the Mozilla Firefox consumer release, the compatibility of the Edge appliance with other versions of Mozilla Firefox cannot be guaranteed.

[2] The Google Chrome support is based on Google Chrome Version 84. Due to frequent version upgrades of Google Chrome, compatibility of the Edge appliance with future versions of Google Chrome cannot be fully guaranteed. Possible incompatibilities will be removed during the regular maintenance process of Cumulocity.

You may also use recent smartphone and tablet web browsers. We have tested our products with the following mobile web browsers:

- Chrome on Android (latest version) on Galaxy smartphones and tablets
- Safari on iOS (latest version) on Apple iPhone and iPad

❗ INFO

Edge on mobile devices shows some limitations. The limitations could be the following:

- The usage may be constrained by the memory and the processing power available on the devices.

For example, loading graphs with large amounts of data points may make the mobile device unresponsive.

- Using the private mode on browsers may not work.
- The [Streaming Analytics application](#) does not support mobile or touch devices.

ACCESSING THE EDGE APPLIANCE USING THE DOMAIN NAME

The Edge appliance is accessible using the domain name configured as part of the installation.

There are two ways to configure the accessibility with the domain names:

- Add an entry of the domain name and IP mapping in the DNS servers.
OR
- [Add the alias](#) to access the Edge appliance through the domain name provided during installation. This must be performed on each client host on which the Edge appliance is accessed.

INFO

The first option is always preferable so that the Edge appliance is accessible over LAN.

Adding the alias

On Linux machines, add the following entry to `/etc/hosts`:

```
<IP address> <domain_name>
```

Use the IP address provided during the network configuration. For example, the default value for Hyper-V is 192.168.66.10.

On Windows machines, add the same entry to `C:\Windows\System32\drivers\etc\hosts`.

Ping the `<domain_name>` to verify it.

```
ping <domain_name>
```

If the ping is successful, the DNS resolution is working properly.

Using `<domain_name>`, the Edge appliance can be connected from the host operating system (operating system which is hosting the Edge appliance). If you want to connect the Edge appliance within your LAN, which is outside of the host operating system, you must do the following:

- On VMware platforms, port forwarding must be enabled as mentioned in [Port forwarding on a VMware platform](#).
- The DNS entry must be added in your LAN's DNS server/Name server. The DNS entry must have the domain name and the IP address of the host operating system. Note that this is not the Edge appliance IP.

To access the Edge appliance

Enter the URL in the browser:

```
https://<domain_name>
```

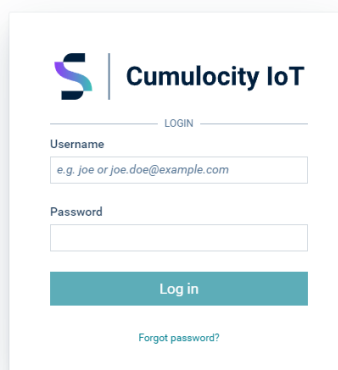
The Edge appliance login screen appears. Log in with your credentials created during the installation.

- To log in to the Management tenant, prefix the username with *management*:
 - Username: `management/<Edge admin username>`
 - Password: password provided during the installation
- To log in to the "edge" tenant, use the Edge admin credentials or prefix the Edge admin username with *edge*:
 - Username: `edge/<Edge admin username>`
 - Password: password provided during the installation

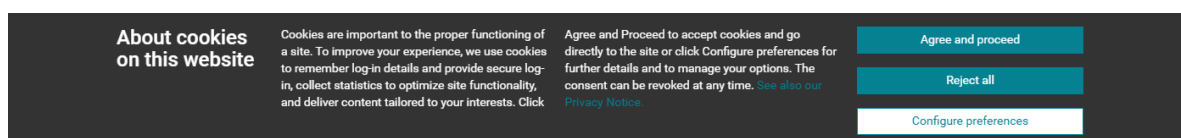
! IMPORTANT

Make sure that the address bar of your browser shows a lock icon. The lock icon indicates that you are using a secure connection and that you are indeed connected to the Edge appliance.

If you are logging in for the first time, you will see a cookie banner at the bottom:



The image shows the Cumulocity IoT login interface. At the top is the Cumulocity IoT logo. Below it is a 'LOGIN' section with a 'Username' field containing the placeholder text 'e.g. joe or joe.doe@example.com', a 'Password' field, and a 'Log in' button. A link for 'Forgot password?' is located below the button.



The image shows a cookie banner at the bottom of the page. It contains the following text: 'About cookies on this website', 'Cookies are important to the proper functioning of a site. To improve your experience, we use cookies to remember log-in details and provide secure log-in, collect statistics to optimize site functionality, and deliver content tailored to your interests. Click', 'Agree and Proceed to accept cookies and go directly to the site or click Configure preferences for further details and to manage your options. The consent can be revoked at any time. See also our Privacy Notice.', and three buttons: 'Agree and proceed', 'Reject all', and 'Configure preferences'.

i INFO

The cookie banner is turned on by default on the Edge appliance instances. This feature can be configured, see [Branding](#).

- Click **Agree and Proceed** to accept the default cookie settings (required and functional cookies enabled).
- Click **Reject all** to reject all of the default cookie settings.
- Click **Preferences** to select your individual cookie preferences:
 - **Required** - Required to enable core site functionality. They perform a task or operation without which a site's functionality would not be possible. Required cookies cannot be disabled.
 - **Functional** - Used to track site usage and to process personal data to measure and improve usability and performance. Functional cookies must be actively enabled.
- Click **See also our Privacy Notice** to open the [Cumulocity privacy statement](#) with details on the Cumulocity privacy policy.

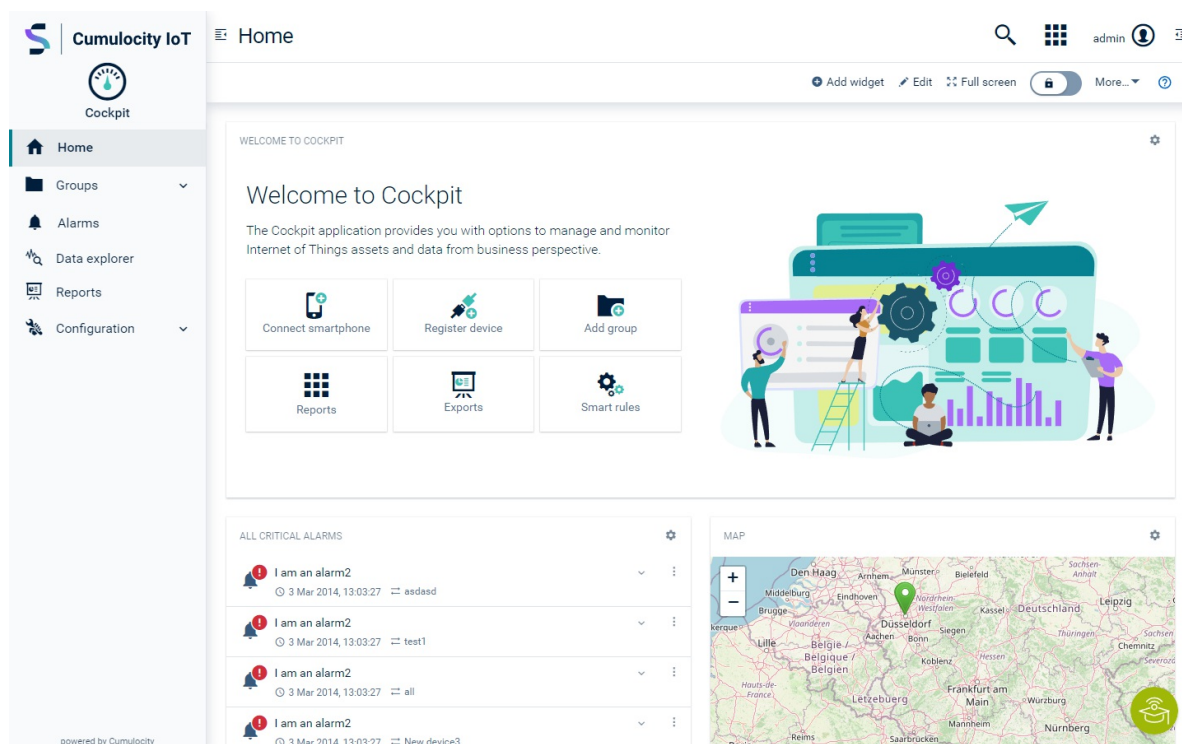
i INFO

If you have enabled functional cookies you can opt-out from the product experience tracking later on via the **User settings** dialog, see [User options and settings](#).

Select the **Remember me** checkbox if you want the browser to remember your credentials, so that you do not have to enter them again

when opening the application the next time. This is especially convenient if you frequently switch between Cumulocity applications, as the Edge appliance requests you to authenticate each time when starting an application. You can make the browser “forget” your credentials by explicitly logging out.

Finally, click **Login** to enter the Edge appliance. Initially, you will be taken to the **Cockpit** application (if not configured differently).



To explicitly logout, click the **User** button at the right of the top bar, then select **Logout** from the context menu.

INFO

The maximum number of failed logins (due to invalid credentials), after which a user is locked, can be configured by the Management tenant on platform level, see *Cumulocity Core - Operations guide*. The default value is 100.

HOW TO RESET YOUR PASSWORD

To reset your password, you must first configure the “reset password” template and email server settings in the Edge appliance. For information about configuring the email server, see [Configuring the email server](#).

For information about resetting the password, see [How to reset your password](#).

HOW TO ACCESS PAGES USING URLS

For information about accessing pages using the URLs, see [How to access pages using URLs](#).

SECURITY HARDEN THE EDGE APPLIANCE

For information about security configuration, see [Configuring security](#).

AVAILABLE APPLICATIONS

See [Available applications](#) for more information.

UI FUNCTIONALITIES AND FEATURES

See [UI functionalities and features](#) for more information.

USER OPTIONS AND SETTINGS

See [User options and settings](#) for more information.

KNOWLEDGE HUB

See [Knowledge Hub](#) for more information.

UPDATING EDGE

This section describes how to update Cumulocity Edge using the GUI and the REST API. An update refers to applying a patch/fix to an existing Cumulocity Edge installation or updating your Edge appliance to a higher version.

Keeping your Edge appliance updated ensures that the Edge appliance is running the latest version with new features and enhancements, and helps in improving the security vulnerabilities and performance of the Edge appliance.

You can update the Edge appliance using [the GUI](#) or [the REST API](#).

UPDATING EDGE USING THE GUI

You can update Edge or apply fixes to your existing installation using the Administration application in the Management tenant.

Download the Edge archive file **Cumulocity Edge (version) Update** from the [Cumulocity Download Center](#).

When you update your Edge appliance, the update also applies operating system patches and fixes for security vulnerabilities. Cumulocity recommends you to create a backup of your existing Edge installation before performing any update.

! IMPORTANT

For information about upgrading from an earlier version to Edge 10.18, refer to [Migrating Edge 10.17 to 10.18](#).

To update Edge:

1. Log in to the Management tenant.
2. Switch to the Administration application using the application switcher at the right of the top bar .
3. Click **Edge > Update** in the navigator.
4. Click **Update > Next**.
5. Upload the new Edge archive file and click **Next**.
6. Click **Update**.

While updating the Edge appliance with the microservice hosting feature enabled, the following notification may appear on the user interface:

Could not get the details on the update status. Do refresh after some time to check if the update has completed.

Refresh the page to check if the update has been completed by verifying the version number.

7. Restart your Edge appliance.

UPDATING EDGE USING THE REST APIS

To update Edge using the REST APIs, see the [edge/update](#) API in the [Cumulocity Edge OpenAPI Specification](#).

MIGRATING EDGE 10.17 TO 10.18

The following section outlines the procedure for migrating from Edge 10.17 to 10.18. Since Edge 10.18 is based on the Rocky Linux

operating system, the upgrade process entails a side-by-side installation accompanied by data migration.

! IMPORTANT

If you are using a version earlier than Edge 10.17 and plan to migrate to Edge 10.18, you must first update your Edge to version 10.17. If you are using a version prior to Edge 10.13, you must first update your Edge to version 10.13 before upgrading to 10.17.

For information about upgrading from an earlier version to Edge 10.17, see:

- [Updating Edge using the GUI](#)
- [Updating Edge using the REST APIs](#)

in the *10.17.0 Cumulocity Edge guide*.

BACKING UP DATA AND CONFIGURATION ON EDGE 10.17

In your Edge 10.17 appliance, back up the MongoDB data, Edge, Cumulocity Agent, and OPC UA configurations.

! IMPORTANT

Before attempting the backup, ensure that there is sufficient disk space to save the backup in your Edge 10.17 appliance.

Perform the following steps as a `root` user on your Edge 10.17 appliance.

1. Run the following commands to unmonitor all services and stop the `opcua-mgmt-service`, `opcua-device-gateway`, `smartrule`, `apama`, `cumulocity-core-karaf` and `mongod` services:

```
monit unmonitor all && \
systemctl stop installation-service opcua-mgmt-service opcua-device-gateway smartrule apama cumulocity-core-karaf mongod
```

2. If you have installed Cumulocity DataHub in the Edge 10.17 appliance, run the following commands to stop the `cdh-console`, `cdh-master` and `cdh-executor` services:

```
service cdh-console stop && \
service cdh-master stop && \
service cdh-executor stop
```

3. Back up the MongoDB data, data lake contents from the DataHub if present, Edge, Cumulocity Agent, and OPC UA configurations using the following command to create the `/opt/edge-1017-backup.tar` file:

```
tar -zcf /opt/edge-1017-backup.tar /opt/softwareag /opt/mongodb /var/lib/cumulocity-agent /usr/edge /opt/opcua
```

4. After creating the `/opt/edge-1017-backup.tar` file, copy it to a network drive or storage location that is accessible from the Edge 10.18 appliance you will create in the next step. Once the backup file is safely stored, shut down the Edge 10.17 appliance to prevent any further changes to the system during the migration process. This step is optional and if not performed, you need to copy it directly into the Edge 10.18 appliance once it is created.

INSTALLING EDGE 10.18

Install the Edge appliance 10.18 following the instructions below.

! IMPORTANT

If you have both the Edge 10.17 and Edge 10.18 appliances on the same host machine, ensure that the IP addresses of both appliances are different.

i INFO

Install the Edge 10.18 appliance with the same domain and license as those of the older Edge Appliance VM you are migrating. For detailed steps, refer to [Installing Edge](#).

1. Refer to [Configuring the Edge infrastructure](#) for detailed instructions on importing the Edge 10.18 appliance.
2. Follow the installation procedure on your Edge 10.18 appliance, ensuring to configure the network settings appropriately. For detailed steps, refer to [Installing Edge](#).

MIGRATING BACKUP DATA TO EDGE 10.18

After installing and configuring the Edge 10.18 appliance, proceed to migrate the data and configurations backed up from Edge 10.17 to the new appliance.

Perform the following steps as a `root` user on your Edge 10.18 appliance:

1. Transfer the backup file `/opt/edge-1017-backup.tar` from the Edge 10.17 appliance to your Edge 10.18 appliance.

! IMPORTANT

Before copying the backup, ensure that there is sufficient disk space available on your Edge 10.18 appliance. For example, if the size of the data disk in the Edge 10.18 appliance is 75 GB and the size of the backup file is 100 GB, you must expand the size of the data disk by an additional 125 GB (total 200 GB, of which 100 GB used by the backup tar file) before copying the backup. For instructions on expanding disk size, refer to [Expanding the disk size](#).

2. Run the following commands to unmonitor and stop the relevant services:

```
monit unmonitor all && \  
systemctl stop installation-service opcua-mgmt-service opcua-device-gateway smartrule apama cumulocity-core-karaf mongod
```

3. Cleanup the MongoDB data before replacing the backup data with the following command:

```
rm -rf /opt/mongodb/*
```

4. Untar the backup file using the following command:

```
tar -xf /opt/edge-1017-backup.tar -C /
```

5. Reboot the Edge appliance with the following command:

```
reboot
```

6. Once the Edge appliance restarts, verify its health by following the instructions in the [Health check](#) section. Ensure that the `monit summary` command returns status *OK* for all services and processes as described in [Using Monit](#) section.
7. Follow the steps outlined in [Updating Edge](#) to finalize the data migration process.

CONFIGURING EDGE 10.18 AFTER DATA MIGRATION

After migrating data to Edge 10.18, proceed to configure the appliance to the same level as Edge 10.17.

1. Follow the instructions in [Configuring the time synchronization](#) to synchronize the time settings.
2. Refer to [Configuring the Network](#) for instructions on configuring the network settings.
3. If you have enabled the microservice hosting feature in the Edge 10.17 appliance, follow the instructions in [Enabling the microservice hosting feature](#) to enable the same in Edge 10.18.
4. If you have installed Cumulocity DataHub in the Edge 10.17 appliance, follow the instructions in [Setting up Cumulocity DataHub Edge](#) to install the same in Edge 10.18. Before you proceed with installing Cumulocity DataHub, change the ownership of the data lake folders using the following command:

```
chown -R systemd-coredump:systemd-coredump /opt/mongodb/cdh-*
```

5. For information about security configuration, refer to [Configuring security](#).

For detailed guidance on configuring the Edge 10.18 appliance, consult [Configuring Edge](#).

SAMPLE SCRIPT TO AUTOMATE THE DATA MIGRATION

Cumulocity provides the `edge-1017-to-1018-data-migration.sh` script as a reference for automating the data migration process. You can customize this script according to your requirements. Click [here](#) to download the script.

! IMPORTANT

Cumulocity does not officially support this migration script. It is provided for reference purposes only.

USING THE SCRIPT

1. Install the Edge 10.18 appliance by following the instructions in [Installing Edge 10.18](#)
2. Download the `edge-1017-to-1018-data-migration.sh` script and copy it to your Edge 10.18 appliance. This script migrates the necessary data including data lake contents from the DataHub.
3. Execute the `edge-1017-to-1018-data-migration.sh` script as a `root` user. Provide the IP address and `root` user password for the Edge 10.17 appliance when prompted. Note that you need to enable root login over SSH on the Edge 10.17 appliance for the script to work properly.
4. After executing the script, reboot the Edge appliance using the following command:

```
reboot
```

5. Once the Edge appliance restarts, verify its health by following the instructions in the [Health check](#) section. Ensure that the `monit summary` command returns status *OK* for all services and processes as described in [Using Monit](#) section.
6. Follow the steps outlined in [Updating Edge](#) to finalize the data migration process.
7. Refer to [Configuring Edge 10.18 after data migration](#) for instructions on configuring the new Edge appliance.

CONFIGURING EDGE

Learn how to configure many different aspects of Cumulocity Edge.

CONFIGURING THE NETWORK

Configuring the network enables you to access the Edge appliance using the domain name. If there are any changes in the network configuration due to an update or changes in the network protocols, you must reconfigure the network of your Edge appliance as described below. After reconfiguring the network, you must update the IP mapping for the domain name with the new IP address in the DNS servers, if you have added an entry in the DNS servers. If you are using an alias, you must update the IP address in the hosts file of your operating system.

IMPORTANT

Do not use the IP addresses from the below blocks in your network configuration. These IP addresses are reserved for Edge internal purpose.

- 10.96.0.0/12 - reserved for Kubernetes service
- 10.88.0.1/16 - reserved for Podman network

CONFIGURING THE NETWORK USING THE UI

1. Log in to the Management tenant using the Edge administrator credentials created during the installation.
 - Username: management/<Edge admin username>
 - Password: password provided during the installation
2. Switch to the Administration application using the application switcher at the right of the top bar .
3. Click **Edge > Network** in the navigator.

The current network configuration for the Edge appliance appears.

4. Click **Edit** to reconfigure the network.
5. Provide the new **IP address** for you network. For example, 192.168.66.10
6. Provide the **Netmask IP** for your network. For example, 255.255.255.0
7. Provide the **Gateway IP** for your network. For example, 192.168.66.1
8. Provide the **DNS** for your network. For example, 8.8.8.8

For DNS, do not use the IP addresses 10.96.0.10 and 127.0.0.1.

9. Provide the **Address range CIDR**. For example, 172.18.0.0/16

The CIDR suffix must be between 0 and 27 inclusive. The default value is 172.16.0.0/15.

The **Address range CIDR** must not overlap with the reserved IP addresses. See [Reserved IP addresses](#).

INFO

If the IP address of the Edge appliance overlaps with the Edge appliance's address range, then you must update the **Address range CIDR**.

CONFIGURING THE NETWORK USING THE REST APIS

To configure the network using the REST APIs, see the [/edge/configuration/network](#) API in the [Cumulocity Edge OpenAPI Specification](#).

TROUBLESHOOTING THE NETWORK CIDR

The network CIDR fails in the following scenarios:

1. The network CIDR unavailable on boot time.

Description: The default network CIDR range is not available in the network. If the network range is already utilized, the network CIDR does not start properly and the Kubernetes cluster startup fails.

2. Invalid network CIDR provided during IP change.

Description: You tried to change the network CIDR of the properly running Edge appliance, but the range is not available in the network.

To troubleshoot these scenarios:

1. Configure the Edge appliance's network and gateway to a different network range. Configuring the Edge appliance to a different network disconnects the Edge appliance from the existing network.
2. Configure the network CIDR to a new non-conflicting value.
3. Configure the Edge appliance's network and gateway back to the network range.

CONFIGURING THE HOSTNAME

The default hostname of the Edge appliance is **iot-edge-server**. You can define a unique hostname for your Edge appliance as required. A unique hostname ensures seamless communication within the network, avoids any hostname conflicts with other network devices and improves the security of your Edge device.

CONFIGURING THE HOSTNAME USING THE UI

1. Log in to the Management tenant using the Edge administrator credentials created during the installation.
 - Username: management/<Edge admin username>
 - Password: password provided during the installation
2. Switch to the Administration application using the application switcher at the right of the top bar .
3. Click **Edge > Hostname** in the navigator.
4. Click **Edit** to change the hostname.
5. Provide the new hostname and click **Save**.

CONFIGURING THE HOSTNAME USING THE REST APIS

To configure the hostname using the REST APIs, see the [/edge/configuration/hostname](#) API in the [Cumulocity Edge OpenAPI Specification](#).

CONFIGURING THE TIME SYNCHRONIZATION

For many use cases, and especially when using Apama, the time inside the Edge appliance must be synchronized with the network. Time synchronization helps in logging events accurately and compare the data from multiple devices.

By default, the **chrony** service is responsible for time synchronization with the host operating system. For Hyper-V, the **chronyd** service is disabled by default for accurate time synchronization.

Once the time synchronization is enabled, you can specify the NTP servers and the interval to trigger the time synchronization.

CONFIGURING THE TIME SYNCHRONIZATION USING THE UI

1. Log in to the Management tenant using the Edge administrator credentials created during the installation.
 - Username: management/<Edge admin username>
 - Password: password provided during the installation
2. Switch to the Administration application using the application switcher at the right of the top bar .
3. Click **Edge > Time synchronization** in the navigator.
4. Use the toggle button to enable the time synchronization.
5. Click **Edit** to specify the NTP servers and the interval to trigger the time synchronization.

CONFIGURING THE TIME SYNCHRONIZATION USING THE REST APIS

To configure the time synchronization using the REST APIs, see the [/edge/configuration/time-sync](#) API in the [Cumulocity Edge OpenAPI Specification](#).

CONFIGURING THE MICROSERVICE HOSTING FEATURE

Microservices are server-side applications which may be used to extend the Edge platform with customer-specific functionality. For more information, see [Microservice SDK](#). When you enable or disable the microservice feature, the Device simulator microservice also gets enabled or disabled. To use the Device simulator, add the **Simulator** permission. For more information, see [Global roles](#).

For more information about Device simulator, see [Working with simulators](#).

INFO

Ensure that you have fulfilled the minimum system requirements: 4 logical CPU cores and 8 GB RAM.

If you want to use the microservice hosting feature, ensure that you do not use these IP ranges in your local network where the Edge virtual machines are configured. When you enable the microservice hosting feature, the Kubernetes system reserves these IP ranges on the Edge instances.

- 10.96.0.0/12

Enabling the microservice hosting feature takes about 10 to 15 minutes to complete.

IMPORTANT

To enable or disable the microservice hosting feature, you must have the “Tenant Manager” role.

INFO

For information regarding any microservices related issues, see [Troubleshooting Microservices](#).

ENABLING OR DISABLING THE MICROSERVICE HOSTING FEATURE USING THE UI

1. Log in to the Management tenant using the Edge administrator credentials created during the installation.

- Username: management/<Edge admin username>
 - Password: password provided during the installation
2. Switch to the Administration application using the application switcher at the right of the top bar .
 3. Click **Edge** > **Microservices** in the navigator.
 4. Use the toggle button to enable the microservice hosting feature.

! IMPORTANT

If you have enabled the remote-connectivity and also the microservice hosting feature, disabling the microservice hosting feature, reconfiguring the network or updating the Edge appliance might result in an alarm in the remote tenant for the Kubernetes network interfaces that are removed or altered.

To deploy a microservice, in the Administration application, navigate to **Ecosystem** > **Microservices**, and click **Add microservice**.

i INFO

The **Add microservice** button will not be available if you have not enabled the microservice hosting feature.

Upload the ZIP file for your microservice application. For more information about deploying a microservice, see [Deploying the "Hello world" microservice](#).

After you have enabled the microservice hosting feature, ensure that the microservices are active and healthy before using them. For more information, see [Microservice runtime](#).

Before disabling the microservice hosting feature, you should unsubscribe from all the microservices that have been uploaded. You can also delete the microservice if you are not planning to enable again and subscribe to the same microservice. For more information about developing and hosting a microservice, see [Microservices SDK](#).

i INFO

Edge appliance will be temporarily non-operational during the operation.

ENABLING OR DISABLING THE MICROSERVICE HOSTING FEATURE USING THE REST APIS

To enable or disable the microservice hosting feature using the REST APIs, see the [/edge/configuration/microservices](#) API in the [Cumulocity Edge OpenAPI Specification](#).

DEPLOYING MICROSERVICES WITH A LOWER MANIFEST VERSION

A microservice specifies an API version in the microservice manifest. Depending on this API version, the microservice runs with all or only a restricted set of Linux kernel capabilities. More precisely, all capabilities are granted to microservices with API version 1 whereas only the capability `CAP_NET_BIND_SERVICE` is granted to microservices with API version 2. For more information, see [Microservice migration to API Version 2](#).

By default, only the Management tenant can upload and subscribe to the microservices with API version 1. To improve the security of the Edge appliance, the minimum API version has been configured to API version 2. Due to the minimum API version configuration, you cannot upload and subscribe to a microservice with API version 1 in the "edge" tenant.

! IMPORTANT

The MLW and the Messaging Service microservices use API version 1. To install the MLW and the

Messaging Service on Edge version 10.15 and later, you must subscribe the “edge” tenant to the **Feature-privileged-microservice-hosting** application before uploading the MLW and the Messaging Service microservices with API version 1.

To upload and subscribe to a microservice with API version 1 in the “edge” tenant:

1. Log in to the Management tenant.
2. Enable the microservice hosting feature, if not already enabled.
3. Go to **Tenants > Subtenants** and click the “edge” tenant.
4. Go to **Applications** and subscribe to the **Feature-privileged-microservice-hosting** application from the **Available applications**.

INFO

It might take up to ten minutes until the “edge” tenant is able to upload and subscribe to privileged microservices.

5. Log in to the “edge” tenant and upload a microservice with API version 1.

CONFIGURING SECURITY

CONFIGURING SECURITY USING THE UI

The Edge appliance is security hardened by default. Furthermore, you can apply specific security configurations to enhance the security of your Edge appliance.

IMPORTANT

For information about deployment security best practices and tenant administration security configuration guidelines, see [Security Hardening Guidelines](#).

To apply the security configuration, follow the steps below:

1. Log in to the Management tenant using the Edge administrator credentials created during the installation.
 - Username: management/<Edge admin username>
 - Password: password provided during the installation
2. Switch to the Administration application using the application switcher at the right of the top bar .
3. Click **Edge > Security** in the navigator.
4. Click **Download configuration** to download a sample JSON syntax for the current configuration. You can use the same JSON file in the POST operation using the REST API.
5. Click **Edit configuration** to edit the configuration.

You can either import a JSON file to prefill the values or enter the values manually.

6. Specify the following parameters:

Configuration	Category	Parameter	Description
Operating system	SELinux	SELinux mode	SELinux secures the operating system from unprivileged actions of confined users. When enabled, SELinux has two modes: permissive and enforcing. permissive: In the permissive mode, SELinux policy is not enforced. The system remains operational and SELinux does not deny any operations but only logs AVC messages, which can be then used for troubleshooting, debugging, and SELinux policy improvements. Each AVC is logged only once in this case. enforcing: In the enforcing mode, it enforces the SELinux policy and denies access based on SELinux policy rules. ! IMPORTANT In the enforcing mode, you cannot access the Edge appliance remotely through VNC. Default value: permissive
		SSH enabled	Enables or disables the SSH functionality of the Edge appliance. By default, the SSH functionality is enabled. Changing the value to false disables the SSH functionality. When disabled, you will not be able to SSH into the Edge appliance, locally or remotely. Default value: True
		Login sessions inactivity timeout (seconds)	The idle duration before a user session is terminated. When configured, the user session terminates after the specified idle duration. Default value: 600 seconds Minimum value: 0

Configuration	Category	Parameter	Description
	Linux Auditing System	Audit logging enabled	The <i>Linux Auditing System</i> helps system administrators create an audit trail, a log for every action on the server. You can track the security-relevant events, record the events in a log file, and detect misuse or unauthorized activities by inspecting the audit log files. Enables or disables Linux Auditing System. A value of true enables and creates an audit log file at <code>/var/log/audit/audit.log</code>. Enabling auditing system increases the consumption of disk space. You can configure the operating system to transfer the audit logs to a centralized logging server by configuring Audisp remote logging server. ! IMPORTANT Once enabled, you cannot disable the auditing system configuration. Default value: False
		Audisp remote logging server	The centralized logging server to transfer the audit logs to. When configured, the operating system audit logs are transferred to the specified centralized logging server.
		Server	The IP address or the hostname of the destination server.
		Port	The port on the destination server.
		Rsyslog remote logging server	The centralized logging server to transfer the local logs to. When configured, the local logs are transferred to the specified centralized logging server.
		Server	The IP address or the hostname of the destination server.
		Port	The port on the destination server.
		Protocol	The protocol (TCP or UDP) used to transfer the logs.
		Login banner	The login banner for your Edge appliance.
		Login banner	The login banner for your Edge appliance.
Kubernetes	Audit policy		Kubernetes audit policy defines the rules about what events should be recorded and what data they should include.

Configuration	Category	Parameter	Description
		Level	The audit level controls what Kubernetes event data is recorded. When configured, the Kubernetes event data is logged to the specified level. Logging more data increases the consumption of disk space. None - do not log events. Metadata - log request metadata (requesting user, timestamp, resource, verb, and so on) but not request or response body. Request - log event metadata and request body but not response body. This does not apply for non-resource requests. RequestResponse - log event metadata, request and response bodies. This does not apply for non-resource requests. Default value: None
		Maximum age (days)	The maximum number of days to retain the old Kubernetes audit log files. A high value has more impact on the disk space. Default value: 30 days Minimum value: 0
		Maximum numbers of log files to retain	The maximum number of Kubernetes audit log files to retain. Setting a value of 0 indicates that there is no restriction on the number of files to retain. A high value has more impact on the disk space. Default value: 10 files Minimum value: 0
		Maximum size (megabytes)	The maximum size in megabytes of the Kubernetes audit log file before it gets rotated. A high value has more impact on the disk space. Default value: 100 MB Minimum value: 0

7. Click **Save**.

CONFIGURING SECURITY USING THE REST APIS

To configure security using the REST APIs, see the [/edge/configuration/security](#) API in the [Cumulocity Edge OpenAPI Specification](#).

CHANGING THE DOMAIN NAME

A key feature of Edge is the ability to operate the platform using a custom domain name. This means that you can configure the Edge appliance to serve you using a domain name of choice, for example *.iot.mycompany.com rather than the default URL of Edge. In addition, you can create subtenants using your domain. These will use <subtenantName>.iot.mycompany.com as their host names.

! IMPORTANT

Before you change the domain name, see [Domain name validation for Edge license key generation](#).

CHANGING THE DOMAIN NAME USING THE GUI

1. Log in to the Management tenant using the Edge administrator credentials created during the installation.

- Username: management/<Edge admin username>
 - Password: password provided during the installation
2. Switch to the Administration application using the application switcher at the right of the top bar  .
 3. Click **Edge > Domain/Certificate** in the navigator.

Review the domain name and the SSL certificate.

4. Click **Edit** below **Domain name** to edit the domain name.
5. Provide the new domain name.

If the existing license and SSL certificate files are compatible with the new domain name, you do not have to upload the license and certificate files.

Click **Update**.

6. If the existing license is not compatible with the new domain name, provide the license file for the new domain name.

If the existing license is compatible with the new domain name, you do not have to upload the license file.

7. If the existing certificate is not compatible with the new domain name, provide the SSL certificate and key files. If you do not have an SSL certificate, select **Generate self signed SSL certificate**.

If the existing certificate is compatible with the new domain name, you do not have to upload the SSL certificate and key files.

8. Click **Save**.

CHANGING THE DOMAIN NAME USING THE REST APIS

To change the domain name using the REST APIs, see the [/edge/configuration/domain](#) API in the [Cumulocity Edge OpenAPI Specification](#).

LICENSE AND CERTIFICATE COMPATIBILITY

The example below describes a scenario where an existing license or certificate is compatible with the new domain name:

- If you have a license for the domain myown.iot.com, you can change the domain to myown.iot.com and any single level subdomain, for example sub.myown.iot.com
- If you have a certificate for the domain myown.iot.com, then you can only set the domain to myown.iot.com. If you have a wildcard certificate like *.myown.iot.com, then you must set the domain name to any single level subdomain of myown.iot.com, that is sub.myown.iot.com, but not myown.iot.com itself.

UPDATING THE SSL CERTIFICATE

You must always have a valid SSL certificate for your domain name that is configured. If the validity of the certificate expires or if you have changed your domain name, you must upload or generate a new SSL certificate. You can upload the certificate using the GUI and REST APIs.

UPDATING THE SSL CERTIFICATE USING THE GUI

1. Log in to the Management tenant using the Edge administrator credentials created during the installation.
 - Username: management/<Edge admin username>
 - Password: password provided during the installation
2. Switch to the Administration application using the application switcher at the right of the top bar  .
3. Click **Edge > Domain/Certificate** in the navigator.

Review the domain name and the SSL certificate.

4. Click **Edit** below **SSL Certificate** to upload the new SSL certificate file and the key file.
5. Provide the new SSL certificate file and the SSL certificate key file.

If you do not have an SSL certificate, select **Generate self-signed certificate** to generate one.

6. Click **Save**.

UPDATING THE SSL CERTIFICATE USING THE REST APIS

To update the SSL certificate using the REST APIs, see the [/edge/configuration/certificate](#) API in the [Cumulocity Edge OpenAPI Specification](#).

CONFIGURING REMOTE CONNECTIVITY

For information about connecting your Edge appliance to the cloud tenant, see [Connecting Edge to the cloud](#).

CONFIGURING KARAF

To enhance the performance and resource allocation for Karaf, you can change the environment variables for Karaf (such as `JAVA_MAX_MEM` and the microservice proxy variables like `MICROSERVICE_RUNTIME_PROXY_HTTP_HOST`, `MICROSERVICE_RUNTIME_PROXY_HTTP_PORT`). To do so, you must update the file `/usr/share/cumulocity-core-karaf/bin/setenv`:

1. Log in to the Edge appliance.
2. Open the file `/usr/share/cumulocity-core-karaf/bin/setenv`.
3. Edit the parameters. If the parameter you want to change is present in the file, update its value; otherwise, add a new line: `export <PARAMETER_NAME>=<VALUE>` at the end of the file.

After changing the file, restart the `cumulocity-core-karaf` service:

```
sudo service cumulocity-core-karaf stop
```

and

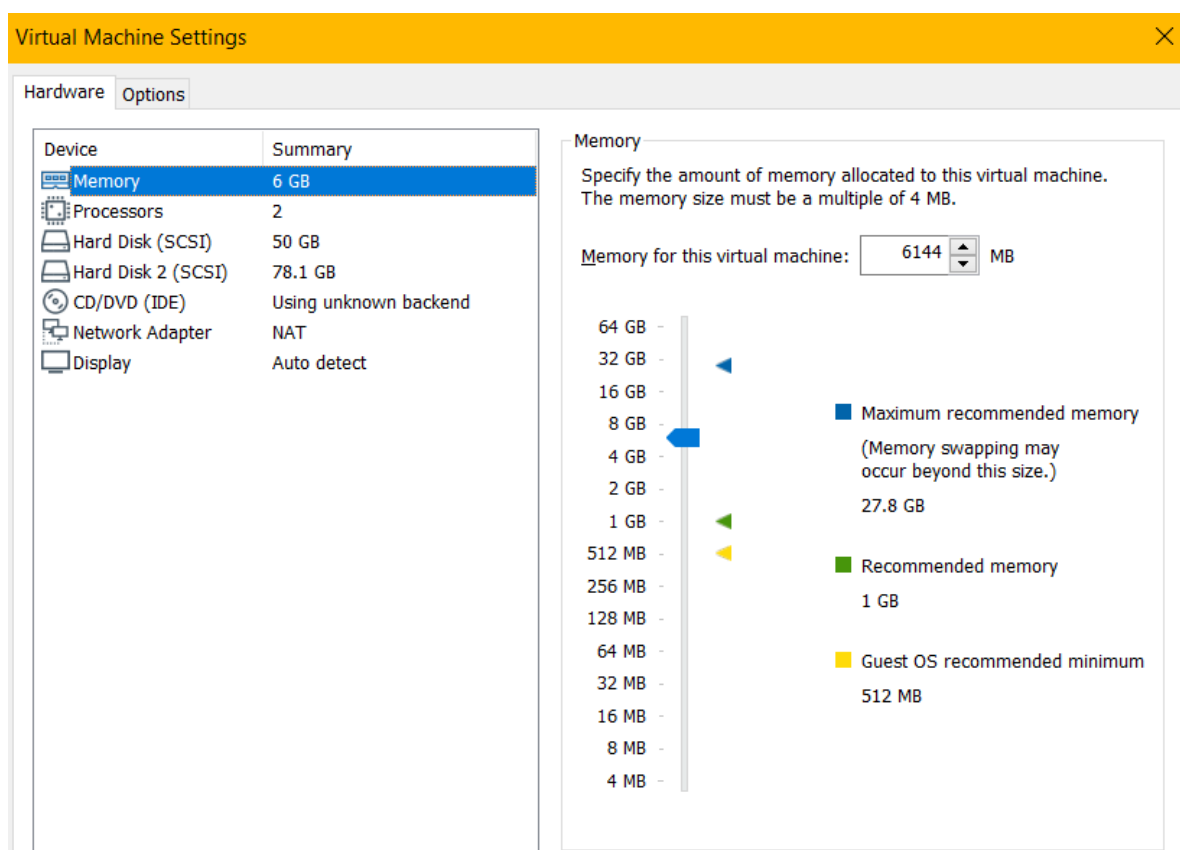
```
service cumulocity-core-karaf start
```

❗ IMPORTANT

Since the changes to this file are overwritten when the Edge appliance is updated, you must reapply the changes to this file after the update process.

INCREASING THE SYSTEM PERFORMANCE

If the system performance is slow, you must increase the memory. Stop the Edge appliance and increase the memory of the Edge appliance using the hypervisor.



Increasing the memory of the Edge appliance must be followed by increasing the memory of the JVM. To increase the memory of the JVM, edit the value of the `JAVA_MAX_MEM` parameter as desired by following the steps described above. The default value of this parameter is 2048 MB.

CHANGING LOG LEVEL FOR KARAF

This section describes how to change the log level for Edge specific applications in the back-end. It does not explain how to change log settings for standard components like databases or other operating system related services.

The log level for Karaf is defined in the following file.

```
/usr/share/cumulocity-core-karaf/etc/org.ops4j.pax.logging.cfg
```

The file has the following structure:

```

# Root logger
log4j.rootLogger=INFO,out,osgi:*
log4j.throwableRenderer=org.apache.log4j.OsgiThrowableRenderer

# Error appender
log4j.appender.out=org.apache.log4j.rolling.RollingFileAppender
log4j.appender.out.rollingPolicy=org.apache.log4j.rolling.FixedWindowRollingPolicy
log4j.appender.out.rollingPolicy.maxIndex=10
log4j.appender.out.triggeringPolicy=org.apache.log4j.rolling.SizeBasedTriggeringPolicy
log4j.appender.out.triggeringPolicy.MaxFileSize=104857600
log4j.appender.out.rollingPolicy.FileNamePattern=${karaf.data}/log/error-%i.log.gz
log4j.appender.out.rollingPolicy.ActiveFileName=${karaf.data}/log/error.log
log4j.appender.out.layout=org.apache.log4j.PatternLayout
log4j.appender.out.layout.ConversionPattern=%d{yyyy-MM-dd} %d{HH:mm:ss} | %-5.5p | %-16.16t | %-32.32c{1} | %X{bundle.id} - %X{bundle.name} - %X{bundle.version} | %m%n
log4j.appender.out.append=true

# CXF request and response info:
# * ERROR - none
# * INFO - just headers (default)
# * DEBUG - whole, with payloads
log4j.additivity.com.cumulocity.rest.interceptors=false
log4j.logger.com.cumulocity.rest.interceptors=INFO,access

# Access appender
log4j.appender.access=org.apache.log4j.rolling.RollingFileAppender
log4j.appender.access.rollingPolicy=org.apache.log4j.rolling.FixedWindowRollingPolicy
log4j.appender.access.rollingPolicy.maxIndex=10
log4j.appender.access.triggeringPolicy=org.apache.log4j.rolling.SizeBasedTriggeringPolicy
log4j.appender.access.triggeringPolicy.MaxFileSize=262144000
log4j.appender.access.rollingPolicy.FileNamePattern=${karaf.data}/log/access-%i.log.gz
log4j.appender.access.rollingPolicy.ActiveFileName=${karaf.data}/log/access.log
log4j.appender.access.layout=org.apache.log4j.PatternLayout
log4j.appender.access.layout.ConversionPattern=%d{yyyy-MM-dd} %d{HH:mm:ss} | %-5.5p | %-16.16t | %-32.32c{1} | %X{bundle.id} - %X{bundle.name} - %X{bundle.version} | %m%n
log4j.appender.access.append=true

# Error response info:
# * INFO - just error message (default)
# * DEBUG - full stack trace
log4j.logger.com.cumulocity.rest.mediatypes=INFO

```

Change the following entries to adjust the log levels:

```

log4j.rootLogger=INFO,out,osgi:*

log4j.logger.com.cumulocity.rest.interceptors=INFO,access

log4j.logger.com.cumulocity.rest.mediatypes=INFO

```

Adjust the log levels by changing the level attribute according to the following values. The levels are inclusive - meaning a given level will also include all "lower" log levels, for example, when you set the level to WARN you will also get ERROR events.

Level	Description
ERROR	Log errors only
WARN	Give information up to warnings
INFO	Give information about normal operations
DEBUG	Log all internal debug information

Save the file. It is re-read by the application every few minutes so you do not have to restart the Java process.

CONFIGURING OPCUA

UPDATING THE MEMORY AND CPU LIMIT FOR OPC UA

The memory limit for OPC UA container is set to 2048 MB and the CPU limit is set to 256 as default values. For most use cases, these values are sufficient. You can change the values as per your requirements.

```
CTRL_DOCKER_OPTIONS="--memory 2048m --cpu-shares 256"
```

You can specify the CPU limit in `--cpu-shares` or `--cpus`.

- `--cpu-shares` specifies the share or percentage of the host machine's CPU cycles allocated. For example, `--cpu-shares 256` allocates a quarter of total CPUs available.
- `--cpus` specifies how much of the available CPU resources the OPC UA container can use. For example, `--cpus 2` allocates two CPUs out of the available CPUs from the host machine.

For more information about CPU limits, see Docker documentation.

To change the memory and CPU limit:

1. Log in to the Edge appliance.
2. Open the files:
 - `/etc/init.d/opcua-device-gateway`
 - `/etc/init.d/opcua-mgmt-service`
3. Edit the parameter `CTRL_DOCKER_OPTIONS` to the required values.
4. Restart the OPC UA services:
 - `sudo service opcua-mgmt-service restart`
 - `sudo service opcua-device-gateway restart`

! IMPORTANT

Since the changes to this file are overwritten when the Edge appliance is updated, you must reapply the changes to this file after the update process.

CONFIGURING APAMA

UPDATING THE MEMORY AND CPU LIMIT FOR APAMA

The memory limit for Apama container is set to 2048 MB and the CPU limit is set to 256 as default values. For most use cases, these values are sufficient. You can change the values as per your requirements.

```
CTRL_DOCKER_OPTIONS="--memory 2048m --cpu-shares 256"
```

You can specify the CPU limit in `--cpu-shares` or `--cpus`.

- `--cpu-shares` specifies the share or percentage of the host machine's CPU cycles allocated. For example, `--cpu-shares 256` allocates a quarter of total CPUs available.
- `--cpus` specifies how much of the available CPU resources the Apama container can use. For example, `--cpus 2` allocates two CPUs out of the available CPUs from the host machine.

For more information about CPU limits, see Docker documentation.

To change the memory and CPU limit:

1. Log in to the Edge appliance.
2. Open the file `/etc/init.d/apama`.
3. Edit the parameter `CTRL_DOCKER_OPTIONS` to the required values.
4. Restart the Apama service:

- `sudo service apama restart`

! IMPORTANT

Since the changes to this file are overwritten when the Edge appliance is updated, you must reapply the changes to this file after the update process.

CONFIGURING THE EMAIL SERVER

Configuring an email server enables you to receive email notifications about events, alarms, and also to reset your password. In case you forget the password, the Edge appliance mails you the password reset link to reset your password.

To configure the “reset password” template and email server settings, perform the following steps:

1. Log into the Management tenant using <https://<tenant-domain>/apps/administration/index.html#/configuration>.
 - Username: `management/<Edge admin username>`
 - Password: password provided during the installation
2. Update the email server details and templates as mentioned in [Password reset](#) and [Email server](#).

CONNECTING EDGE TO THE CLOUD

Take a look at how to configure and manage your Cumulocity Edge appliance remotely.

CONFIGURING REMOTE CONNECTIVITY USING THE UI

1. Log in to the Management tenant.
 - Username: management/<username>
 - Password: password provided during the installation
2. Switch to the Administration application using the application switcher at the right of the top bar .
3. Click **Edge > Remote Connectivity** in the navigator.
4. Switch the **Enable remote connectivity** toggle to enable remote-connectivity.
5. Provide the URL of the Cumulocity cloud tenant in the **Remote tenant URL** field.
6. Click **Save**.

Next steps: Note down the Edge device ID and [register your Edge appliance](#) in the Cumulocity cloud tenant.

CONFIGURING REMOTE CONNECTIVITY USING THE REST APIS

To configure remote connectivity using the REST APIs, see the [edge/configuration/remote-connectivity](#) API in the [Cumulocity Edge OpenAPI Specification](#).

REGISTERING THE EDGE APPLIANCE IN THE CUMULOCITY TENANT

Before performing these steps, ensure that you have [configured the URL](#) for the Cumulocity cloud tenant in the Edge appliance.

1. Log in to your Cumulocity tenant.
2. Go to the Device Management application.
3. Click **Registration** in the **Devices** menu and then click **Register device**.
4. Select **General device registration**.



Register a general device

Device ID

Add to group

e.g. 0123ab32fcd (require)

No items

+ Add device

Cancel

Next

5. In the **Device ID** field, enter the Edge device ID.
6. Click **Next**.
7. Click **Complete** to register your Edge appliance.
After successful registration, the Edge appliance appears in the [Device registration](#) page with the status **Waiting for connection**.
Turn on the Edge appliance and wait for the connection to be established. Once the device is connected, the device status changes to **Pending acceptance**.
8. Click **Accept** to confirm the connection. The status of the device changes to **Accepted**.

ACCESSING THE EDGE APPLIANCE FROM THE CUMULOCITY TENANT

The Cumulocity Cloud Remote Access allows you to remotely access the Edge appliance through a web browser. The remote Edge appliance is represented as a device in the Device Management application of Cumulocity.

PREREQUISITES

To use Cloud Remote Access, you need:

- “Remote access” permission granted to the tenant user.
- A Cumulocity tenant subscribed to the Cloud Remote Access microservice. To get the subscription, contact [product support](#).

SUPPORTED PROTOCOLS

The following protocols are supported to connect to the Edge appliance through remote access from the Cumulocity tenant:

- Remote Desktop (VNC). See [Accessing the Edge device remotely through VNC](#).
 - Shares the desktop of the remote device
 - Mouse and keyboard for interaction
- Secure Shell (SSH)
 - Console for command line access
 - Keyboard for interaction

❗ IMPORTANT

Starting with Edge Release 10.15, the support for the Telnet protocol has been removed, as Telnet is

considered to be an insecure protocol lacking built-in security measures. For accessing the Edge appliance remotely from the Cumulocity tenant, Cumulocity recommends you to use the SSH protocol instead.

For more information about remote access, see [Cloud Remote Access](#).

ACCESSING THE EDGE APPLIANCE REMOTELY THROUGH VNC

You can access the Edge appliance from the Cumulocity tenant by installing the VNC components on your Edge appliance. Ensure that you have registered your Edge appliance with the Cumulocity tenant. See [Registering the Edge appliance in the Cumulocity tenant](#).

INFO

VNC server does not work if SELinux is set to enforcing mode.

STEP 1: INSTALLING THE VNC COMPONENTS

INFO

The Edge appliance must be connected to the internet to install the VNC components.

1. Run the script `vnc-setup.sh`.

```
sudo /opt/c8y/utilities/vnc-setup.sh
```

The `vnc-setup.sh` script installs the VNC components. After installing the VNC components, you should configure the VNC server for each user.

2. After installing the VNC components, log in as a root user and add the `vnc-server` service to the `firewalld` firewall. For example, the following commands add the `vnc-server` to the `firewalld` with the zone set to public:

```
firewall-cmd --zone=public --permanent --add-service=vnc-server
```

```
firewall-cmd --reload
```

INFO

These are sample commands and may vary based on your requirements.

STEP 2: CONFIGURING VNC SERVER FOR A USER

To configure the VNC server, run the `vnc-user-setup.sh` script. Each user should run this script to set up VNC components and be able to connect to the Edge appliance. The `vnc-user-setup.sh` script enables the current user to use VNC functionality and set a VNC password for the current user.

1. Run the script `vnc-user-setup.sh`.

```
/opt/c8y/utilities/vnc-user-setup.sh
```

2. Provide and verify the password.

3. Select **Y** or **N** to enter a view-only password.

Record the allocated port number. This port number will be used to connect to the VNC server on your Edge appliance.

i INFO

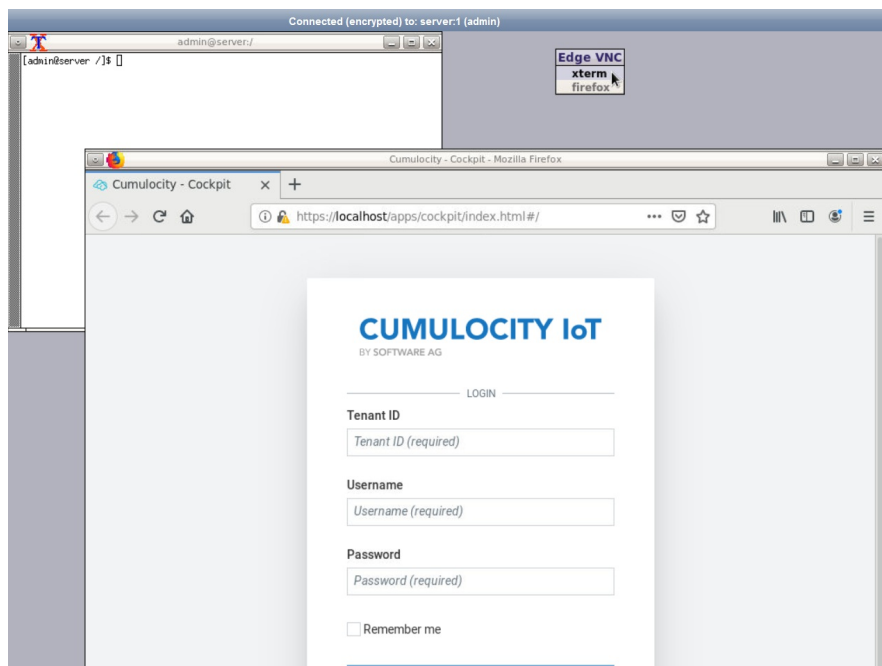
You can also get the allocated port number from the `/opt/c8y/utilities/vnc-display-mapping` file. In this file, you will find the VNC display number allocated for each user. For example, `admin:1`. You must add 5900 to the number associated with the user. In this example, the port number for the user **admin** is 5901.

STEP 3: CONNECTING TO THE EDGE APPLIANCE USING VNC

To access and connect to the Edge appliance:

1. In the Cumulocity tenant for your registered Edge appliance, add a remote access endpoint. See [To add a remote access endpoint via VNC](#). You must use the same port number that is allocated for you.
2. Connect to the endpoint. See [To connect to an endpoint](#).

The connection to the Edge appliance is established and the UI appears for the Edge appliance. Right-click in the screen to open the desktop components **xterm** and **firefox**.



i IMPORTANT

Do not use the **Send Ctrl+Alt+Del** button in the VNC interface. If you do so, you will lose the VNC connection and not be able to reconnect until you restart the Edge appliance.

CHANGING THE VNC PASSWORD

You can change the VNC password for the current user by running the `vnc-user-setup.sh` script. After changing the password, you must update the password in the remote access endpoint.

ACCESSING THE EDGE APPLIANCE REMOTELY THROUGH SSH

You can access the Edge appliance from the Cumulocity tenant through SSH.

INFO

Ensure that you have registered your Edge appliance with the Cumulocity tenant. See [Registering the Edge appliance in the Cumulocity tenant](#).

To access and connect to the Edge appliance:

1. In the Cumulocity tenant for your registered Edge appliance, add a remote access endpoint. See [To add a remote access endpoint via SSH](#). If you select the **Sign-in method** as **Username and password**, you must enter the operating system's administrator credentials. By default, the port number should always be 22.
2. Connect to the endpoint. See [To connect to an endpoint](#).

RESTARTING EDGE REMOTELY

The Cumulocity tenant allows you to remotely restart your Edge appliance. To do so:

1. Log in to your Cumulocity tenant.
2. Go to the Device Management application.
3. Click **Devices** > **All devices** in the navigator.
4. Select the Edge appliance that you want to restart.
5. Click **Restart device**.



UPDATING EDGE REMOTELY

You can remotely update your Edge appliance using the Cumulocity's firmware update feature. This process requires you to upload the Edge archive file (tar.gz) to your tenant account or a server (you must provide the location of the archive on that server). For more information about the firmware feature, see [Managing firmware](#).

IMPORTANT

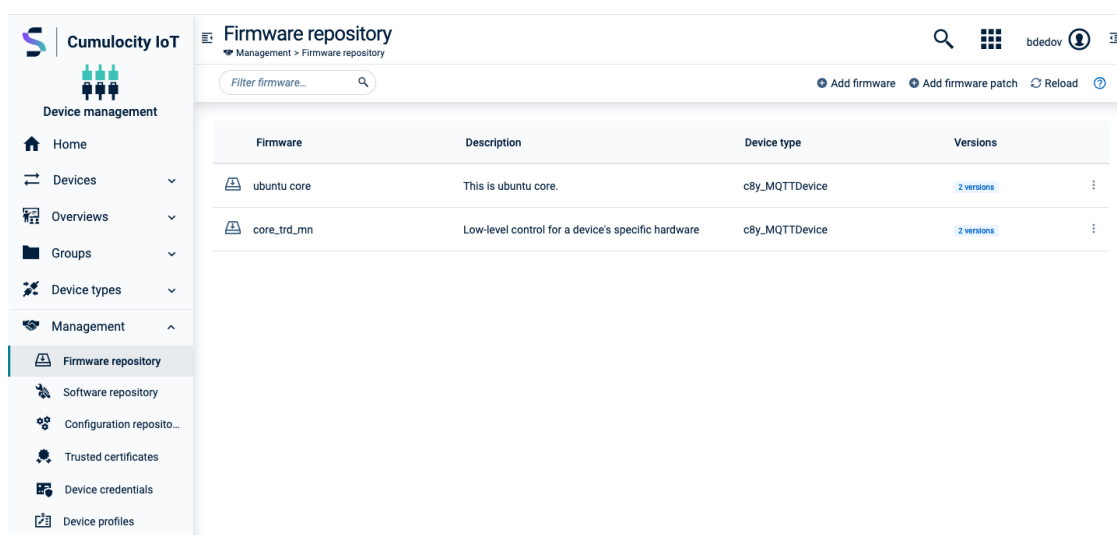
The Edge archive file size is about 3GB. If your tenant account has a limitation for uploading files with size over 1GB, contact [product support]/[additional-resources/contacting-support/ to increase the maximum file size for upload.

Before you update your Edge appliance remotely, you must:

- [Enable the remote-connectivity](#) feature in your Edge appliance
- [Register your Edge appliance](#) in your Cumulocity tenant account

To remotely update your Edge appliance:

1. Log in to your Cumulocity tenant account.
2. Go to the Device Management application.
3. Click **Firmware repository** in the **Management** menu in the navigator.



4. Click **Add firmware** at the right of the top menu bar.
5. Provide a name for the firmware, add a description and its version (all required).
6. Optionally, you can define the device type filter when adding a new firmware. For example, *c8y_EdgeAgent*.
7. Select the **Provide a file path** option to specify an HTTPS URL of a server from where the archive file can be downloaded.
8. Click **Add firmware**. The firmware object appears in the firmware list.
9. Click **All devices** in the **Devices** menu in the navigator, select your Edge appliance from the device list.
10. Click **Firmware**. The **Firmware** tab shows the current Edge appliance version.

The screenshot displays the CUMULOCITY IoT Device Management web interface. On the left is a blue sidebar with the logo and a list of navigation options: Home, Devices (selected), Registration, All devices, Map, Simulators, Service monitoring, Overviews, Groups, Device types, and Management. The main content area has a header with the device ID 'edge-agent-fe0051dd-932c-438c-b92c-48cd8df12eb2'. Below the header is a horizontal menu with tabs: Info, Measurements, Alarms, Control, Firmware (selected), Events, Remote access, Service monitoring, and Identity. The 'Firmware' tab is active, showing a section titled 'Current firmware' with a download icon, the text 'VERSION 10.11.0.0.79', and a blue button labeled 'Replace firmware'.

You can also see the current Edge appliance version in the **Info** tab.

Info

Measurements

Alarms

Control

Firmware

Events

Remote access

Service monitori...

Identity

DEVICE DATA

Type	c8y_EdgeAgent
Last updated	2021-10-13T21:06:36.894Z
Creation time	2021-10-13T20:35:47.310Z

AVAILABILITY

Status	AVAILABLE
Last message	2021-10-13T21:19:36.707Z

CONNECTION

Status	CONNECTED
--------	-----------

FIRMWARE

Version	10.11.0.0.79
---------	--------------

REQUIRED AVAILABILITY

Response interval	20
-------------------	----

Takes a value between -32768 and 32767 minutes (a negative value indicates that the device is under maintenance).

11. Click **Replace firmware**.
12. Select the firmware that you just uploaded and click **Install**.

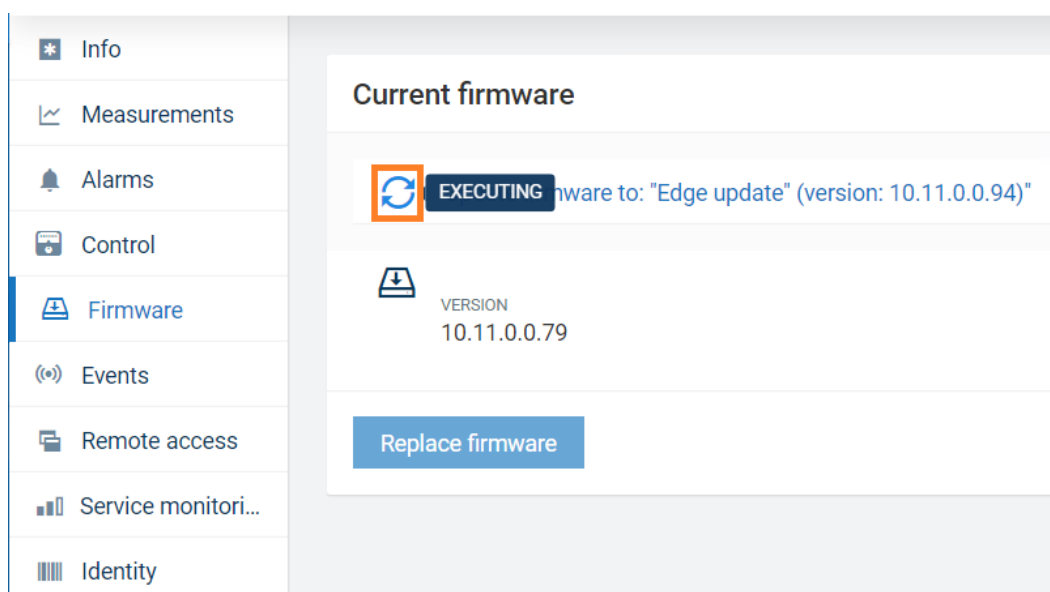
Edge update

Edge update 94

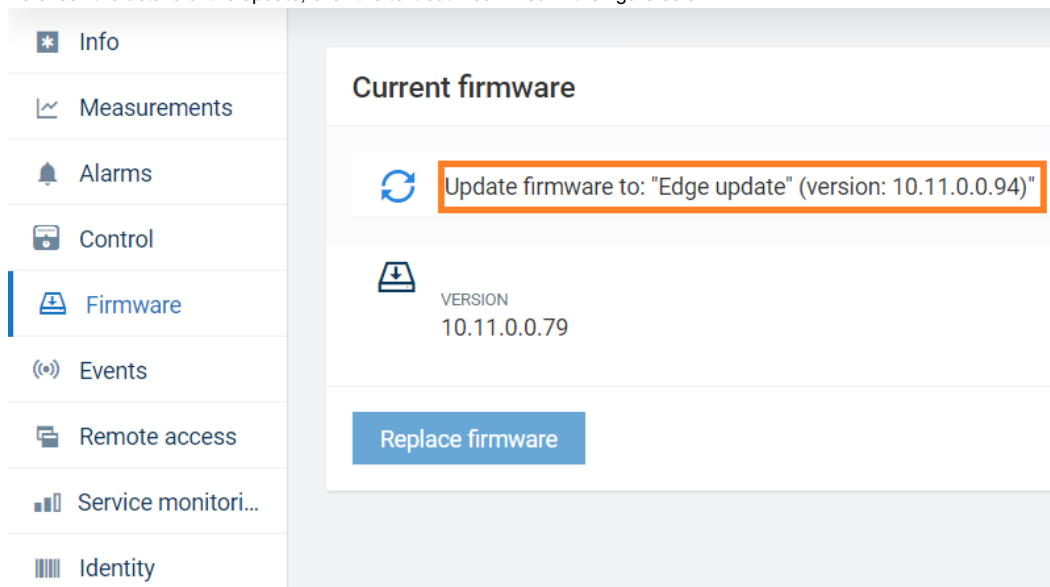
1 version

☒ 10.11.0.0.94

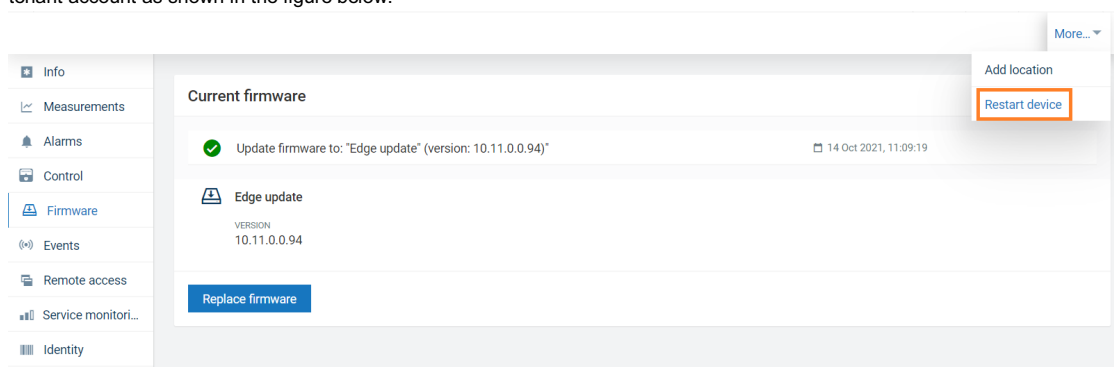
- To check the status of the update, hover over the refresh icon  as shown in the figure below:



- To check the details of the update, click the text outlined in red in the figure below:



- Restart your Edge appliance after the firmware update is complete. You can remotely restart your Edge appliance from your tenant account as shown in the figure below:



UPDATING EDGE APPLIANCES USING BULK OPERATIONS

In your tenant account, you can update multiple Edge appliances using the bulk operations feature. To do so, follow the steps below:

1. Log in to your Cumulocity tenant account.
2. Go to the Device Management application.

3. Click **Device control** in the **Overviews** menu in the navigator.

CUMULOCITY IoT DEVICE MANAGEMENT

Home, Devices, **Overviews**, Alarms, Device control, Events, Groups, Device types, Management

Device control

Single operations | **Bulk operations**

Type: All | All statuses | Filter by date

Realtime | New bulk operation | Reload

Operation	Progress	Status
4 Update firmware to: "Edge_update" (version: 1.00) START 12 Oct 2021, 22:49:00 FINISH 12 Oct 2021, 22:49:01	100%	
3 Update firmware to: "edgetest" (version: 1.0) START 5 Oct 2021, 21:50:00 FINISH 5 Oct 2021, 21:50:02	100%	2 successful
2 Update firmware to: "edgetest" (version: 1.0) START 5 Oct 2021, 18:35:00 FINISH 5 Oct 2021, 18:35:02		
1 Update firmware to: "edgetest" (version: 1.0) START 5 Oct 2021, 15:05:00 FINISH 5 Oct 2021, 15:05:02	100%	2 successful

4. Click **Bulk operations** and then click **New bulk operation**.

CUMULOCITY IoT DEVICE MANAGEMENT

Home, Devices, **Overviews**, Alarms, Device control, Events, Groups, Device types, Management

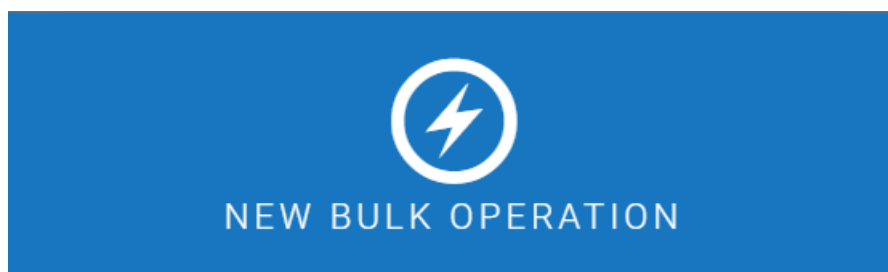
Device control

Single operations | **Bulk operations**

Type: All | All statuses | Filter by date

Realtime | **New bulk operation** | Reload

Operation	Progress	Status
4 Update firmware to: "Edge_update" (version: 1.00) START 12 Oct 2021, 22:49:00 FINISH 12 Oct 2021, 22:49:01	100%	
3 Update firmware to: "edgetest" (version: 1.0) START 5 Oct 2021, 21:50:00 FINISH 5 Oct 2021, 21:50:02	100%	2 successful
2 Update firmware to: "edgetest" (version: 1.0) START 5 Oct 2021, 18:35:00 FINISH 5 Oct 2021, 18:35:02		
1 Update firmware to: "edgetest" (version: 1.0) START 5 Oct 2021, 15:05:00 FINISH 5 Oct 2021, 15:05:02	100%	2 successful



Select one of the available options

Check the [User guide](#) for more details.

5. Click **Firmware update**.

Configuration update

Firmware update

Software update

Apply device profile

Cancel

6. Select a firmware from the list. The list can be filtered by firmware name. Click **Next**.
7. Expand the version and select the archive file. Click **Next**.

Select version or patch

Edge update
Edge update 94

Version	File
10.11.0.0.94	CumulocityIoTEdge-10.11.0.0.94.tar.gz

8. Select the Edge appliances that you want to update by applying filters to the paginated list of all devices.

You can filter by status, name, type, model, group, registration date and alarms. You may apply multiple filters. To apply a filter, click the column header, make your filter option choices in the context menu and click **Apply**.

Select target devices 3265 of 3268 items Clear filters

Status	Name	Type	Model	Group	Registration date	Alarms
OK	edge-agent-2236e770-e488-4e0c-b54c-9cd7ecd37e55	c8y_EdgeAgent			14 Oct 2021, 11:38:56	2
OK	edge-agent-0c6ae4ea-c50b-4bfc-9521-c9311d705d9e	c8y_EdgeAgent			14 Oct 2021, 11:09:01	1
OK	edge-agent-8326f0f9-3bec-4d69-a442-550e342a7aaa	c8y_EdgeAgent			14 Oct 2021, 10:50:51	1

Cancel Back Next

9. Enter a new title or use the preset title. Optionally enter a description. Select a start date and a delay. The delay may either be in seconds or milliseconds and is the time spent between each single operation of the bulk operation. Click **Schedule bulk operation** to create the bulk operation.

✓

Select firmware

✓

Select version or pat...

✓

Select target devices

4

Confirm and schedul...

Confirm and schedule bulk operation

Update firmware to: Edge update (version: 10.11.0.0.94).

Description

Firmware update for Edge appliances with device type c8y_EdgeAgent

Start date

10/14/2021

14

:

12

Delay

1

Seconds

Cancel

Back

Schedule bulk operation

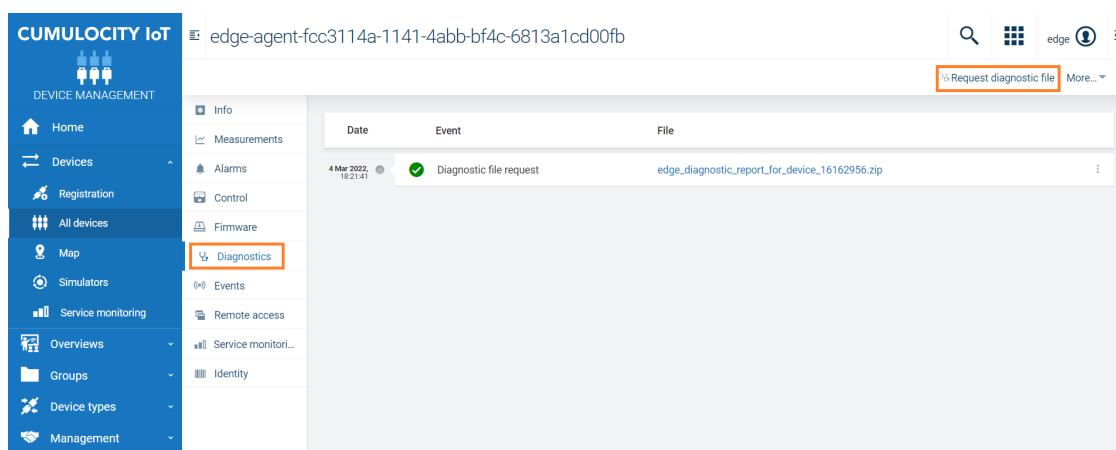
For more information about bulk operations, see [Working with operations](#).

DOWNLOADING DIAGNOSTICS REMOTELY

The Cumulocity tenant allows you to download the diagnostics report remotely from your tenant account after you have registered your Edge appliance in the tenant account.

To download the diagnostics report from your tenant account:

1. Log in to your Cumulocity tenant.
2. Go to the Device Management application.
3. Click **Devices** > **All devices** in the navigator.
4. Select the Edge appliance for which you want to download the diagnostics report.
5. Click **Diagnostics** > **Request diagnostic file**.



6. Click the ZIP file link to download the diagnostic report.

DATA EXCHANGE USING DATA BROKER

Data broker lets you upload the data to a Cumulocity tenant account selectively. Note that you must first create a Cumulocity tenant account.

! IMPORTANT

Data broker and Edge connectivity are compatible only with the current version and previous two releases of Cumulocity. Cumulocity has a release every three months.

You can share the following data with the tenant account:

- Devices (and more generically, managed objects)
- Events
- Alarms
- Measurements

! IMPORTANT

Data broker in Edge does not support synchronization of the [operations](#).

To upload the data to a Cumulocity tenant account, you must first create a data connector in the Edge appliance and subscribe this connector in the tenant account.

A data connector describes the subset of the data that you would like to send to a destination tenant. For more information, see [Data connectors](#).

To create a data connector and upload the data to the tenant account, perform the following steps:

1. In your Edge appliance, log in to the "edge" tenant.
2. In your Edge appliance, go to the Administration application. Click **Data broker > Data connectors**.
3. Click **Add data connector** and provide all the information and filters. See [To add a data connector](#).

Note down the security code. This security code will be used to subscribe the connector in the tenant account.

4. Log in to the Cumulocity tenant account.
5. In the tenant account, go to the Administration application. Click **Data broker** > **Data subscriptions** to subscribe the connector created in your Edge appliance.

Click **Add data subscription** and provide the security code. Click **Submit** and accept the subscription. See [Data subscriptions](#).

You can now navigate to the Device Management application or the Cockpit application. You will find a new “virtual group” with a specific icon showing the forwarded devices. The group will have the same name as your subscription.

For more information about sending and receiving data in Cumulocity, see [Using the data broker](#).

CONNECTING DEVICES TO EDGE

INTRODUCTION

Edge uses REST and MQTT as protocols for external communication. Both protocols may be used to interface devices with Edge. For more information, see [Device integration using REST](#) and [SmartREST 2.0](#).

Additionally, Edge offers:

- Cloud Fieldbus functionality to collect data from fieldbus devices and remotely manage them. For example, Modbus protocol.
- OPC UA protocol. OPC UA protocols support through the OPC UA device gateway and OPC UA management service.

For details on how to integrate devices using Modbus and OPC UA protocols, see [Connecting a Modbus device](#) and [Connecting an OPC UA device](#).

INFO

Currently, only the Modbus and OPC UA protocols are supported.

CONNECTING A MODBUS DEVICE

Modbus is a serial communications protocol originally published by Modicon and used to establish primary-replica/client-server communication between devices.

Before you connect a Modbus device, ensure that the agent is running.

HOW TO CHECK/CHANGE THE AGENT STATE

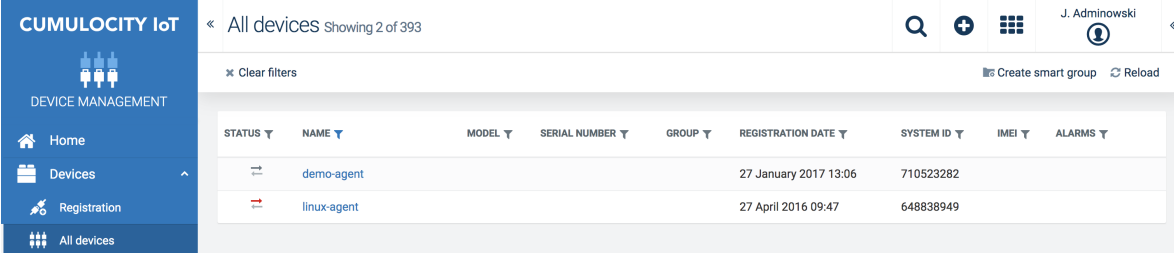
Use the following command to check if the agent is running in Edge (running on port 6670):

```
systemctl status cumulocity-agent
```

If the agent is not running, start it with the following command:

```
systemctl start cumulocity-agent
```

The Modbus agent is pre-registered. In the Device Management application, click **All devices** in the navigator and find the Modbus agent (called "linux-agent") in the device list.



STATUS	NAME	MODEL	SERIAL NUMBER	GROUP	REGISTRATION DATE	SYSTEM ID	IMEI	ALARMS
	demo-agent				27 January 2017 13:06	710523282		
	linux-agent				27 April 2016 09:47	648838949		

HOW TO CONNECT MODBUS DEVICES

For more information about connecting and managing Modbus devices, see [Cloud Fieldbus](#).

CONNECTING AN OPC UA DEVICE

! IMPORTANT

While configuring the OPC UA server, ensure that the server is reachable from the Edge appliance. If you are using a hostname, configure the DNS accordingly.

To connect an OPC UA device in the Edge appliance, follow the steps below.

PREPARATION

OPC UA Management service

Check if the OPC UA management service is running in Edge (running on port 8083):

```
sudo service opcua-mgmt-service status
```

If the agent is not running, start it with the following command:

```
sudo service opcua-mgmt-service start
```

OPC UA Device Gateway

Check if the OPC UA device gateway is running in Edge (running on port 1099):

```
sudo service opcua-device-gateway status
```

If the agent is not running, start it with the following command:

```
sudo service opcua-device-gateway start
```

REGISTERING THE DEVICE

Next, you must register a device in the Device Management application with the name `opcua-gateway`.



REGISTER DEVICES

DEVICE INFO

REGISTRATION

Device ID

Add to group

Select group ▼

 Add another device

Back

Next

Follow the description in [Registering devices](#) to register a device.

In the Device Management application, click **All devices** in the navigator and find the OPC UA device in the device list.

For further information about managing and configuring OPC UA devices, see [OPC UA](#).

EDGE OPERATIONS

This section describes the main operating procedures for standard tasks that must be carried out when managing Cumulocity Edge.

RESTARTING THE EDGE APPLIANCE

When you apply a patch or update your Edge appliance to a newer version, you must restart your Edge appliance after the update. Before restarting your Edge appliance, ensure that your appliance is in a safe state.

INFO

You might see alarms when you power on the Edge appliance from hibernation.

RESTARTING THE EDGE APPLIANCE USING THE UI

To restart your Edge appliance:

1. Log in to the Management tenant using the Edge administrator credentials created during the installation.
 - Username: `management/<Edge admin username>`
 - Password: password provided during the installation
2. Switch to the Administration application using the application switcher at the right of the top bar .
3. Click **Edge > Reboot** in the navigator.
4. Click **Reboot**.

RESTARTING THE EDGE APPLIANCE USING THE REST APIS

To restart the Edge appliance using the REST APIs, see the `edge/reboot` API in the [Cumulocity Edge OpenAPI Specification](#).

STARTING AND STOPPING SERVICES

IMPORTANT

If the processes are monitored by **monit**, the processes restart after you run the `stop` command.

KARAF PROCESSES

In the Edge appliance, run the following commands as admin user.

Process	Command
Start	<code>sudo service cumulocity-core-karaf start</code>

Process	Command
Stop	<code>sudo service cumulocity-core-karaf stop</code>

OPCUA-MGMT-SERVICE

In the Edge appliance, run the following commands as admin user.

Process	Command
Start	<code>sudo service opcua-mgmt-service start</code>
Stop	<code>sudo service opcua-mgmt-service stop</code>
Restart	<code>sudo service opcua-mgmt-service restart</code>
Status	<code>sudo service opcua-mgmt-service status</code>

OPCUA-DEVICE-GATEWAY

In the Edge appliance, run the following commands as admin user.

Process	Command
Start	<code>sudo service opcua-device-gateway start</code>
Stop	<code>sudo service opcua-device-gateway stop</code>
Restart	<code>sudo service opcua-device-gateway restart</code>
Status	<code>sudo service opcua-device-gateway status</code>

SMARTRULE

In the Edge appliance, run the following commands as admin user.

Process	Command
Start	<code>sudo systemctl start smartrule</code>
Stop	<code>sudo systemctl stop smartrule</code>
Restart	<code>sudo systemctl restart smartrule</code>

APAMA

In the Edge appliance, run the following commands as admin user.

Process	Command
Start	<code>sudo service apama start</code>
Stop	<code>sudo service apama stop</code>

Process	Command
Restart	<code>sudo service apama restart</code>

EXPANDING THE DISK SIZE

You can expand the disk size of the installation disk and the data disk using the UI and REST APIs. You can either expand the disk size for both the disks or any one of the disk at a time. There is no limit on the number of the disk expansion process. Before expanding the disk size, you must set or edit the disk size in the hypervisor. See the hypervisor specific documentation for editing the disk size.

EXPANDING THE DISK SIZE USING THE UI

1. Shut down your Edge appliance.
2. Increase the size of the installation and data disks in you hypervisor.
3. Restart your Edge appliance.
4. Log in to the Management tenant using the Edge administrator credentials created during the installation.
 - Username: `management/<Edge admin username>`
 - Password: password provided during the installation
5. Switch to the Administration application using the application switcher at the right of the top bar .
6. Click **Edge > Expand disk size** in the navigator.
7. Click **Expand**.

EXPANDING THE DISK SIZE USING THE REST APIS

To expand the disk size using the REST APIs, see the [edge/expand-disk](#) API in the [Cumulocity Edge OpenAPI Specification](#).

HEALTH CHECK

NETWORK

The following instructions show how to check the network connectivity of the platform.

Check network interface of the node

The following commands will show the interface and network settings of the machine:

```
ip a
```

This will list all interfaces and its current configuration.

Example:

```
[admin@iot-edge-server ~]$ ip a

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
    valid_lft forever preferred_lft forever

2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
link/ether 08:00:27:88:e7:de brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 brd 10.0.2.255 scope global noprefixroute dynamic enp0s3
    valid_lft 85338sec preferred_lft 85338sec
inet6 fe80::a00:27ff:fe88:e7de/64 scope link noprefixroute
    valid_lft forever preferred_lft forever

3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
link/ether 08:00:27:81:fe:9d brd ff:ff:ff:ff:ff:ff
inet 192.168.56.120/24 brd 192.168.56.255 scope global noprefixroute enp0s8
    valid_lft forever preferred_lft forever
inet6 fe80::5b3a:bc65:40b5:f9ea/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
```

You must make sure that the node has an external interface (ethX) and the loopback interface configured (lo). The loopback interface must have the fixed IP 127.0.0.1 with subnet mask 255.0.0.0, and the IP address of the external interface must reside in the correct subnet with the correct subnet mask (in this example 255.255.252.0).

The following command lists the local routing information.

```
netstat -rn
```

Example:

```
[admin@iot-edge-server ~]$ netstat -rn
Kernel IP routing table
Destination Gateway Genmask Flags MSS Window irtt Iface
0.0.0.0 10.0.2.2 0.0.0.0 UG 0 0 0 enp0s3
10.0.2.0 0.0.0.0 255.255.255.0 U 0 0 0 enp0s3
192.168.56.0 0.0.0.0 255.255.255.0 U 0 0 0 enp0s8
```

Make sure you have the destination 0.0.0.0 in the list which then also has the gateway flag (G) set.

Check access to the internet

Try to reach a well-known address in the internet with the following command:

```
ping -s 1500 8.8.8.8
```

Example: [admin@iot-edge-server ~]\$ ping -s 1500 8.8.8.8 PING 8.8.8.8 (8.8.8.8) 1500(1528) bytes of data. 64 bytes from 8.8.8.8: icmp_seq=1 ttl=56 time=2.61 ms 64 bytes from 8.8.8.8: icmp_seq=2 ttl=56 time=2.80 ms 64 bytes from 8.8.8.8: icmp_seq=3 ttl=56 time=2.82 ms 64 bytes from 8.8.8.8: icmp_seq=4 ttl=56 time=2.75 ms 64 bytes from 8.8.8.8: icmp_seq=5 ttl=56 time=2.79 ms

As when checking the internal reach ability, you must make sure that you can see the replies from the address you tried to reach. Use Ctrl-C to end the ping command.

PROCESSING

This section lists the required services and processes on the Edge appliance.

Check platform status

You can check the status of the platform by running the following command:

```
curl -v http://localhost:8181/tenant/health

* About to connect() to localhost port 8181 (#0)
*   Trying ::1...
* Connected to localhost (::1) port 8181 (#0)
> GET /tenant/health HTTP/1.1
> User-Agent: curl/7.29.0
> Host: localhost:8181
> Accept: */*
>
< HTTP/1.1 200 OK
< Content-Type: application/json
< Date: Sat, 05 May 2018 18:13:28 GMT
< Transfer-Encoding: chunked
< Server: Jetty(8.1.19.v20160209)
<
* Connection #0 to host localhost left intact
{
  "status": "UP",
  "services": {
    "details": {},
    "status": "UP"
  },
  "mongodb": {
    "details": {},
    "status": "UP"
  },
  "mongodb-cluster": {
    "details": {
      "mongodb-cluster-enabled": true
    },
    "noSuppressibleDown": false,
    "status": "UP"
  },
  "tenant": {
    "details": {},
    "status": "UP"
  }
}
```

The 'status:"UP"' shows that all services are running.

If something fails, the endpoint should respond a different error code. The response should contain something like:

```

{
  "status":"DOWN",
  "services":{
    "details": {},
    "status":"UP"
  },
  "mongodb":{
    "details": {},
    "status":"UP"
  },
  "mongodb-cluster": {
    "details": {
      "mongodb-cluster-enabled": true
    },
    "noSuppressibleDown": false,
    "status": "UP"
  },
  "tenant":{
    "details":{
      "notFullyInitializedTenants":{
        "savenindia":"PAYPAL_REFUND",
        "nowa":"PAYPAL_REFUND",
        "ttc":"PAYPAL_REFUND",
        "controllogichh":"PAYPAL_REFUND",
        "jonathan":"PAYPAL_REFUND",
        "barcotrail":"PAYPAL_REFUND",
        "ngservices":"PAYPAL_REFUND",
        "sixsenses":"PAYPAL_REFUND",
        "trav2":"PAYPAL_REFUND",
        "lucaslocatrix":"PAYPAL_REFUND"
      }
    },
    "status":"DOWN"
  }
}

```

The response contains checks for the most important components:

- services - status of OSGI services running within karaf container
- mongo - status of connection to mongo database (for clustered mongo shows status of connection to mongos)
- mongodb-cluster - status of mongo cluster, status DOWN doesn't have bad impact for overall health status (because it is marked as "noSuppressibleDown": false) Mongo-cluster shows status as follow:
 - "details": {"mongodb-cluster-enabled": true}, "status": "UP" - mongo cluster is enabled and works correctly
 - "details": {"mongodb-cluster-enabled": false}, "status": "DOWN" - mongo cluster is disabled and mongo works in single mode (for health status check section "mongo")
 - "details": {"mongodb-cluster-enabled": true}, "status": "DOWN" - mongo cluster is enabled and works incorrectly, additionally mongo exception is presented
- tenant - tenant initialization status

The list describes the tenants which on the core node have not been fully initialized and the initialization status they are in.

Status	Description
UNINITIALIZED	Tenant has not been initialized
DB_INITIALIZED	Database initialisation is performed
LOCAL_APPLICATION_INITIALIZED	Local applications are deployed
CEP_INITIALIZED	CEP rules are been deployed
HEART_BEAT_MONITORING_INITIALIZED	Heartbeat monitoring for devices is been started

Status	Description
PAYPAL_REFUND	Failed PayPal operations are refunded (when tenant is using PayPal)
BULK_OPERATION_INITIALIZED	Support for bulk operations is initialized
FULLY_INITIALIZED	Tenant is in working state (should not be displayed)

CHECKING THE SYSTEM SERVICES

Checking REST API availability

Run the following command to check the REST API availability:

```
curl -u 'edge/<username>:<password>' -v -X GET http://<base_url>/platform
```

```
* About to connect() to <base_url> port 80 (#0)
* Trying 52.29.189.245... connected
* Connected to <base_url> (52.29.189.245) port 80 (#0)
* Server auth using Basic with user 'management/<username>'
> GET /platform HTTP/1.1
> Authorization: Basic bWFuYWdlbWVudC90c3NjaHVlbDohITQ3TmV1bjl3MQ==
> User-Agent: curl/7.19.7 (x86_64-redhat-linux-gnu) libcurl/7.19.7 NSS/3.21 Basic ECC zlib/1.2.3 libidn/1.18 libssh2/1.4.2
> Host: <base_url>
> Accept: */*
>
< HTTP/1.1 200 OK
< Server: nginx
< Date: Tue, 23 Aug 2016 15:39:29 GMT
< Content-Type: application/vnd.com.nsn.cumulocity.platformApi+json; charset=UTF-8; ver=0.9
< Transfer-Encoding: chunked
< Connection: keep-alive
<
{"alarm":{"alarms":{"alarms":null,"self":"http://management.<base_url>/alarm/alarms"},"alarmsForSource":"http://management.<base_url>/alarm/alarms?source={source}","alarmsForSourceAndStatus":"http://management.<base_url>/alarm/alarms?source={source}&status={status}","alarmsForSourceAndStatusAndTime":"http://management.<base_url>/alarm/alarms?source={source}&status={status}&dateFrom={dateFrom}&dateTo={dateTo}","alarmsForSourceAndTime":"http://management.<base_url>/alarm/alarms?source={source}&dateFrom={dateFrom}&dateTo={dateTo}","alarmsForStatus":"http://management.<base_url>/alarm/alarms?status={status}","alarmsForStatusAndTime":"http://management.<base_url>/alarm/alarms?status={status}&dateFrom={dateFrom}&dateTo={dateTo}","alarmsForTime":"http://management.<base_url>/alarm/alarms?dateFrom={dateFrom}&dateTo={dateTo}","self":"http://management.<base_url>/alarm"},"..."}
```

This example shows the correct response of the platform. The username and password must have full read access to the Management tenant. The <base_url> must be given to connect to the correct platform and the Management tenant must not be blocked from outside.

MONITORING

Monit is a small open-source utility for managing and monitoring the Unix systems. The Monit utility conducts automatic maintenance and repair, and can execute meaningful causal actions in error situations.

In Edge, Monit is used to monitor the processes and perform a restart if any of the processes is down or inactive. The usual Monit interval (cycle) is 30 seconds.

USING MONIT

Monit can provide a quick status report of all the configured services and processes by running the following command:

```
sudo monit summary
```

There might be cases where Monit has stopped monitoring some resources because of timeout on constant failures or dependency issues.

```
[admin@iot-edge-server ~]$ sudo monit summary
Monit 5.30.0 uptime: 35m
```

Service Name	Status	Type
iot-edge-server	OK	System
ssl_management_proc	OK	Process
smartrule_proc	OK	Process
opcua_device_gateway_proc	OK	Process
opcua_mgmt_service_proc	OK	Process
nginx_proc	OK	Process
mongod_standalone	OK	Process
karaf_proc	OK	Process
cumulocity-agent_proc	OK	Process
apama-ctrl_proc	OK	Process
ssl_management_rc	OK	File
smartrule_rc	OK	File
opcua_device_gateway_rc	OK	File
opcua_mgmt_service_rc	OK	File
nginx_bin	OK	File
openresty_rc	OK	File
mongo	OK	File
mongod	OK	File
karaf_rc	OK	File
nginx_nofile	OK	Program
mqtt_status	OK	Program
server_mongodb	OK	Program
microservice_hosting	OK	Program

A specific component, for example, `apama-ctrl_proc`, can be restarted using the following command:

```
sudo monit restart apama-ctrl_proc
```

The status of a specific component, for example, `apama-ctrl_proc`, can be viewed using the following command:

```
sudo monit status apama-ctrl_proc
```

The Monit status can be checked by running:

```
sudo systemctl status monit
```

Monit can be restarted by running:

```
sudo systemctl restart monit
```

The log file for monit is located in `/var/log/monit.log`.

MONITOR

The monitor can be started by running the `run_monitor.py` script located under `/opt/c8y/utilities/diagnostic-utility/src/`.

The monitor script supports only one optional startup parameter:

`-s` or `--skipDataCollector` allows you to skip the data collection even if one or more monitored components are not working.

DIAGNOSTIC UTILITY

The diagnostic utility is a script to collect, for example, the journal logs and performance metrics of various components running on the Edge appliance, which are essential for you and the Cumulocity Support team to troubleshoot the problems. The diagnostic utility is enabled by default and is scheduled to run periodically. However, you can also trigger it manually using the command line (below) or the [user interface](#).

```
cd /opt/c8y/utilities/diagnostic-utility
sudo ./run_data_collector.py
```

HARDWARE INFORMATION

The basic hardware information of the target system is captured. These reports are placed under the 'hardware' directory.

The following hardware information is available:

Information	Description
CPU	Reads and gathers CPU information from <code>"/proc/cpuinfo"</code>
Memory	Reads and gathers memory information from <code>"/proc/meminfo"</code>
Detailed system summary	Data is captured using the 'lswm' command. Data includes multiple components like network, display adapter, bridge, IDE, and so on. This command is executed as sudo to capture all available details.
Short system summary	Data is captured using the 'lswm' command, here the data is in precise format. This command is executed as sudo to capture all available details.
PCI	Data related to installed PCI devices is captured using the 'lspci' command
Storage	Data is an aggregation of the output of the commands 'df' and 'lsblk'

SOFTWARE INFORMATION

The basic software information of the target system is captured. These reports are placed under the 'software' directory.

The following software information is available:

Information	Description
IP	Reads basic IP information from the target system using the 'ip' command

Information	Description
OS	Collects various OS information like name, version, release, and so on, using the commands 'lsb_release' and 'uname'
Installed package	A list of installed packages is prepared using the 'rpm' command
Running processes	A list of running processes is prepared using the 'ps' command
Top result	Captures the output of top command. This report is very informative as it holds information of running processes at argument level and their respective resource consumption.

CUMULOCITY INFORMATION

This section contains information on the running Cumulocity processes, health endpoint check result, Cumulocity logs, and so on.

The following Cumulocity information is collected:

Information	Description
Health endpoint result	Cumulocity and its microservices provide health endpoints, from which the user can get the system status.
Mongo command execution result	MongoDB supports commands execution, which can give the status of the MongoDB server. Currently 'ping', 'dbstats' and 'serverStatus' commands are executed on each of the MongoDB nodes (currently it is management and edge). The MongoDB commands give vital information about the MongoDB server like the db version, process-id, uptime information, and so on.
Mongo top output	The output of mongo top command is captured here
Thread dumps	Thread dumps of all the running java processes and mongo processes are captured. For java processes the 'jstack' command is executed to get the thread dumps. For non-java processes like MongoDB, the 'pstack' command is used. Furthermore the 'pstack' command is applied on java processes as well.
Log files	Archive of the log files from Cumulocity, its microservices and Apama is created.
Configuration files	Archive of Cumulocity configuration files from Cumulocity and its microservices is created.
Jstat dumps	Jstat command provides performance statistics for a given JVM.

Jstat dumps can provide information on the following options:

- class - Statistics on the behavior of the class loader
- compiler - Statistics of the behavior of the HotSpot Just-in-Time compiler
- gc - Statistics of the behavior of the garbage collected heap
- gccapacity - Statistics of the capacities of the generations and their corresponding spaces
- gccause - Summary of garbage collection statistics (same as -gcutil), with the cause of the last and current (if applicable) garbage collection events
- gcnnew – statistics of the behavior of the new generations
- gcnnewcapacity - Statistics of the sizes of the new generations and its corresponding spaces
- gcold - Statistics of the behavior of the old and permanent generations
- gcoldcapacity - Statistics of the sizes of the old generations
- gcpermcapacity - Statistics of the sizes of the permanent generations
- gcutil – Summary of garbage collection statistics
- printcompilations – HotSpot compilation method statistics

As per the current implementation, the 5 statistics counts are collected at an interval of 50ms. To improve performance, this task is executed via threads.

OPTIONAL STARTUP PARAMETERS

Following are the supported startup parameters for the monitor and the data collector.

Data collector

The data collector can be started by running the "run_data_collector.py" script located under "/opt/c8y/utilities/diagnostic-utility/".

Following are the supported command line arguments which can be used while invoking the script. More than one of the supported arguments can be used simultaneously.

- -hw or --hardware: Allows the script to collect only the hardware information
- -sw or --software: Allows the script to collect only the software information
- -c8y or --cumulocity: Allows the script to collect only the Cumulocity information
- -h: Displays the help message

MICROSERVICES LOG FILE LOCATIONS

The logs of the Kubernetes components are captured at: */tmp/diagnostic-utility/diagnostic_report_XXXXXX/cumulocity/log_archive/kubernetes_logs.zip*.

The kubernetes_logs.zip file contains the logs of all Kubernetes platform components at "kube-system" path in the archive. The components captured are:

- heapster-XX
- kube-apiserver-server
- kube-controller-manager-server
- kube-dns-XX
- kube-flannel-ds-XX
- kube-proxy-XX
- kube-scheduler-server

INFO

The XX represents randomly generated alphanumeric sequences in these pod names and would vary in your environment.

The hosted microservices are captured at *cumulocity-single-node* path in the archive. The pre-installed component **kube-registry-persistent-secure-xx-xx** is already available in the archive. The logs of any additional microservices that are uploaded will also be available at this path.

UTILITY CONFIGURATION FILE

The diagnostic utility can be customized using a properties file located under "/etc/diagnostic-utility/diagnostic_utility.properties".

IMPORTANT

The SMTP properties in the table below are only for collecting diagnostics information. For configuring the email server, see [Email server](#).

Following are the available keys used in the configuration file:

Information	Description
email.notification	Allows users to select whether they want to receive the diagnostic report via email
recipient.email	Recipient email ID that receives the support email
smtp.server.host	SMTP host for sending support email
smtp.server.port	SMTP port to be used by the utility while sending support email
smtp.username	SMTP username to be used by the utility while sending support email
smtp.password	SMTP password to be used by the utility while sending support email
components.for.log.backup	Components for which the log backup must be performed
components.for.configuration.backup	Components for which the configuration backup must be performed
report.directory	Report directory where the diagnostic reports must be placed
{component-name}.log.path	Absolute log path of the components under "component.for.log.backup" which do not use /var/log as the logging directory
containers.to.monitor	Containers for which log backup is required. The container names are separated by a comma. These log files are retrieved using <code>docker logs</code> .
services.to.monitor	Services for which log backup is required. The service names are separated by a comma. These log files are retrieved using <code>journalctl</code> .

DOWNLOADING DIAGNOSTICS USING THE UI

1. Log in to the Management tenant using the Edge administrator credentials created during the installation.
 - Username: management/<Edge admin username>
 - Password: password provided during the installation
2. Switch to the Administration application using the application switcher at the right of the top bar .
3. Click **Edge > Diagnostics** in the navigator.
4. Click **Request diagnostics report** to download the diagnostics.

To download the diagnostics report remotely, see [Downloading diagnostics remotely](#).

DOWNLOADING DIAGNOSTICS USING THE REST APIS

To download the diagnostics report using the REST APIs, see the [edge/diagnostics](#) API in the [Cumulocity Edge OpenAPI Specification](#).

LOGGING

The solution stores log files at the following locations for the different nodes.

CORE NODE LOG FILE LOCATIONS

Directory	Files	Usage
/var/log/cumulocity	access.log	logfile for client access
	error.log	logfile for errors and general information
	karaf.log	logfile for the Karaf Container
	mqtt.log	logfile for mqtt protocol communication
/var/log/nginx	access.log	logfile for client access
	error.log	logfile for nginx errors
/var/log	messages	general log file, contains also the messages from HAProxy

APAMA LOG FILE LOCATIONS

To access the apama-ctrl log files, run the command:

```
sudo docker logs apama-ctrl-edge
```

MONGODB LOG FILE LOCATIONS

Directory	Files	Description
/var/log/mongodb	mongod.log	Log file generated by mongod service
	mongod.log.X.gz	Archive of previous log files generated by mongod service
	mongomongod7.log	Log file generated by mongodmongod7 service
	mongomongod7.log.X.gz	Archive of previous log files generated by mongodmongod7 service
/var/log	messages	general log file

AGENT LOG FILE LOCATIONS

Component	Files	Location
opcua-mgmt-service	opcua-mgmt-service.log	/var/log/opcua/
opcua-device-gateway	opcua-device-gateway.log	/var/log/opcua/
Smartrule-agent-server-apama	smartrule-agent-server-apama-gc.log smartrule-agent-server-apama.log smartrule.log	/var/log/smartrule/
cumulocity-agent	cumulocity-agent.log	/var/log/cumulocity-agent/

DATAHUB LOG FILE LOCATIONS

See [Log files](#) for details on DataHub log files.

LOG ROTATION

The components for which log rotation is configured are as follows:

Component	Log file location	Log file rotation	Max file size	Max backup index
Karaf	\${karaf.data}/log/error.log	Daily	50 MB	14
MQTT	\${karaf.data}/log/mqtt.log	Daily	50 MB	14
Access	\${karaf.data}/log/access.log	Daily	50 MB	14
DataBroker	\${karaf.data}/log/databroker.log	Daily	50 MB	14

Under */etc/* they are configured via *logrotate.conf* and the configuration files under */etc/logrotate.d*.

The components for which log rotation is configured are as follows:

Component	Log file location	Log file rotation	Max file size	Max backup index
MongoDB	/var/log/mongodb/*.log	Daily	50 MB	14
NginX	/var/log/nginx/*.log	Daily	50 MB	14
Apama	/var/log/apama/*.log	Daily	50 MB	14

For microservices, there currently is no specific log rotation configured.

TROUBLESHOOTING THE SYSTEM

In case of any issues, Cumulocity recommends you to follow these steps:

- Perform a health check, see [Health check](#).
- Monitor the system processes, see [Monitoring](#).
- Check the log files, see [Logging](#).

These steps help you analyze the issue and provide a fix. If you need to contact Cumulocity Support, include the output of the diagnostics dump. For more information about using the diagnostics, see [Diagnostic utility](#).

TROUBLESHOOTING MICROSERVICES

In case of any microservices related issues, we recommend you to check if `_microservice_hosting_` is disabled using the command `sudo monit status microservice_hosting`

```
[admin@iot-edge-server ~]$ sudo monit status microservice_hosting
Monit 5.38.0 uptime: 33m

Program 'microservice_hosting'
  status          OK
  monitoring status Monitored
  monitoring mode  active
  on reboot        start
  last exit value  0
  last output      kubectl command not found.
                  Microservice Hosting is disabled.
  data collected   Tue, 06 Feb 2024 13:36:38
```

If it shows up as disabled,

enable the microservice hosting feature through [the GUI](#) or [the REST API](#). If enabling the microservice hosting feature fails, check the status of the Kubernetes system pods using the command below:

```
sudo kubectl get pods --namespace kube-system
```

If any pod in the list show its status as evicted, the most probable cause is that the system ran out of disk space. Expand the disk size through [the GUI](#) or [the REST API](#). Enable the microservice hosting feature after disk expansion.

When running any kubectl command (for example `sudo kubectl get pods --namespace kube-system`), if the message `-bash: kubectl: command not found` is seen then the microservice hosting feature has not been enabled or the process to enable the microservice hosting feature has failed. Attempt to enable the microservice hosting feature through [the GUI](#) or [the REST API](#).

BACKUP AND RESTORE

Read about the steps to backup and restore an Edge appliance in various environments.

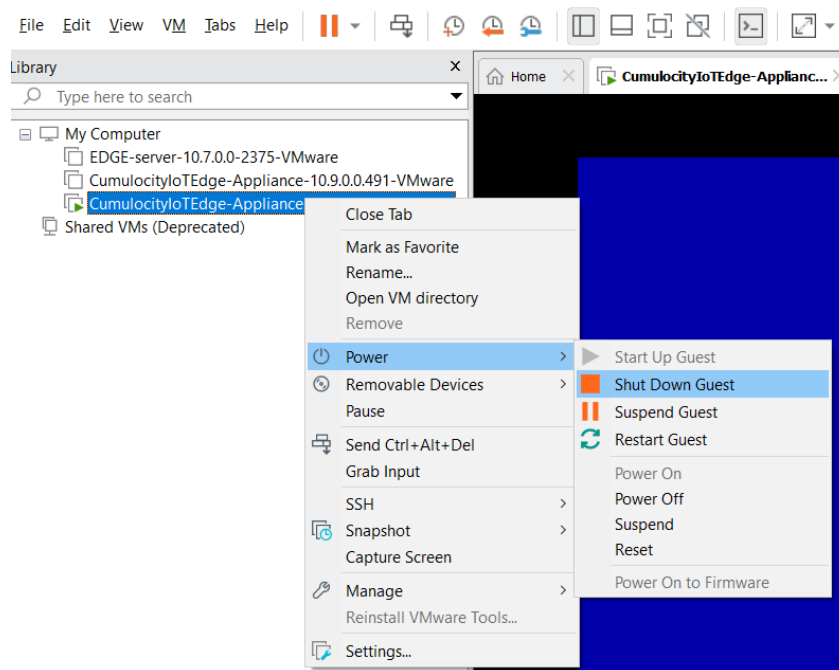
BACKUP AND RESTORE IN VMWARE WORKSTATION PLAYER

VMware Workstation Player does not support backup functionality. However, you can restore a virtual machine in VMware Workstation Player. To restore a virtual machine, go to **Player > File > Open** and follow the prompts.

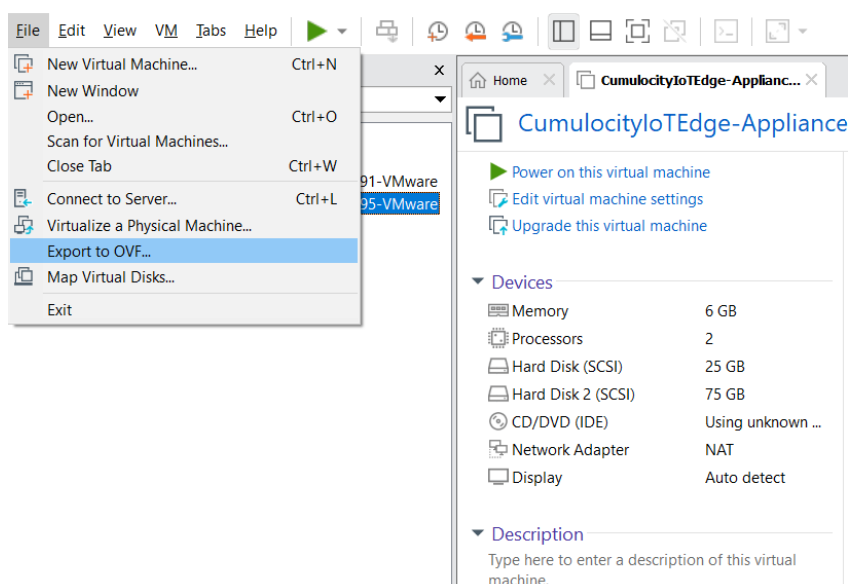
BACKUP AND RESTORE IN VMWARE WORKSTATION PRO

CREATING A BACKUP IN VMWARE WORKSTATION PRO

1. Shut down the Edge appliance.



2. Select the Edge appliance that you want to back up.
3. Click **File > Export to OVF**.



4. Click **Save**.

RESTORING AN EDGE APPLIANCE IN VMWARE WORKSTATION PRO

1. Click **File > Open**.
2. Select the Edge appliance that you want to restore and click **Open**.
3. Click **Import**.

BACKUP AND RESTORE IN VMWARE ESXI

CREATING A BACKUP IN VMWARE ESXI

1. Shut down the Edge appliance in the ESXi console.
2. Select the Edge appliance and click **Actions > Export**.
3. Click **OK** to download the files.

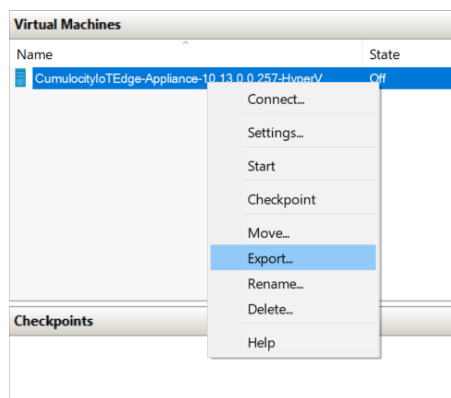
RESTORING AN EDGE APPLIANCE IN VMWARE ESXI

Restoring an Edge appliance from an OVF file is similar to deploying a virtual machine in VMware ESXi. To restore an Edge appliance, perform the steps as described in [Example setup for VMware ESXi](#).

BACKUP AND RESTORE IN HYPER-V

CREATING A BACKUP IN HYPER-V

1. Shut down the Edge appliance in the Hyper-V console.
2. Right-click the Edge appliance and click **Export**.



3. Provide the location to save the backup files and click **Export**.

RESTORING AN EDGE APPLIANCE IN HYPER-V

Restoring an Edge appliance is similar to deploying a virtual machine in Hyper-V. To restore an Edge appliance, perform the steps as described in [Example setup for Hyper-V](#).

USING EDGE FEATURES

BRANDING

With the Branding feature, you can fully customize the look of your tenant to your own preferences. To access the Branding feature:

Log in to the Management tenant using the Edge administrator credentials created during the installation:

- Username: management/<Edge admin username>
- Password: password provided during the installation

Switch to the Administration application using the application switcher at the right of the top bar  .

Go to **Settings** > **Enterprise tenant** and follow the instructions detailed in [Branding](#).

MONITORING THE EDGE METRICS

In your Cumulocity cloud tenant, you can monitor the measurements of the Edge appliance listed in the table below. To monitor the measurements from your cloud tenant, ensure that you have registered your Edge appliance with the Cumulocity cloud tenant. See [Registering the Edge appliance in the Cumulocity tenant](#) for more information.

Measurement	Metrics	Description
Disk space	- Total disk space - Free disk space - Used disk space - Percentage of used disk space	The Edge appliance sends the disk space metrics as a measurement for both installation disk and data disk, every 10 minutes. The measurements are sent in gigabytes (GB) rounded to two decimal places. The percentage is rounded to one decimal place. The data points for this measurement are: - <i>c8y_InstallationDisk</i> - <i>c8y_DataDisk</i> If Edge is unable to read the metrics from the installation disk or the data disk, an alarm is sent to the Cumulocity tenant. The alarms have a minor severity and the data points for the alarms are: - <i>c8y_FileSystemMeasurementErrorInstallationDisk</i> - <i>c8y_FileSystemMeasurementErrorDataDisk</i>

Measurement	Metrics	Description
Memory (RAM)	- Total RAM - Free RAM - Used RAM - Percentage of RAM used	The Edge appliance sends the memory usage metrics as a measurement every 5 seconds in gibibytes (GiB). The data point for this measurement is <i>c8y_Memory</i> If Edge is unable to read the metrics from the memory, an alarm is sent to the Cumulocity tenant. The data point for the alarm is: - <i>c8y_MemoryMeasurementError</i>.
CPU	Percentage of CPU used Unit: Percentage	The Edge appliance sends the percentage of CPU used at intervals over 5 seconds, 60 seconds, and 600 seconds. The data points for this measurement are: - <i>c8y_CpuUsage5Seconds</i> - <i>c8y_CpuUsage60Seconds</i> - <i>c8y_CpuUsage600Seconds</i> If Edge is unable to read the metrics from the CPU, an alarm is sent to the Cumulocity tenant. The data point for the alarm is: - <i>c8y_CPUMeasurementError</i>.
Disk I/O	- Data read per second - Data written per second Unit: KB/s	The Edge appliance sends the disk input/output metrics as a measurement for both installation disk and data disk at intervals over 5 seconds, 60 seconds, and 600 seconds. The data points for this measurement are: - <i>c8y_DataDiskIo5Seconds</i> - <i>c8y_DataDiskIo60Seconds</i> - <i>c8y_DataDiskIo600Seconds</i> - <i>c8y_InstallationDiskIo5Seconds</i> - <i>c8y_InstallationDiskIo60Seconds</i> - <i>c8y_InstallationDiskIo600Seconds</i> If Edge is unable to read the metrics from the disk, an alarm is sent to the Cumulocity tenant. The data point for the alarm is: - <i>c8y_DiskIOMeasurementError</i>.
Network	- Data and packets sent per second - Data and packets received per second Unit: KB/s and packets/s	The Edge appliance sends the network metrics as a measurement at intervals over 5 seconds, 60 seconds, and 600 seconds. The data points for this measurement are: - <i>c8y_NetworkInterface_Io-5Seconds</i> - <i>c8y_NetworkInterface_Io-60Seconds</i> - <i>c8y_NetworkInterface_Io-600Seconds</i> If Edge is unable to read the metrics from the network, an alarm is sent to the Cumulocity tenant. The data point for the alarm is: - <i>c8y_NetworkIoMeasurementError</i>.

To monitor the metrics in your Cumulocity tenant, you can create a dashboard and add widgets in the Cockpit application of your tenant. For more information about creating dashboards, see [Working with dashboards](#).

Also, you can define smart rules to create alerts or raise alarms for the metrics. For example, when the free disk space is less than 5 GB, create an alert. For more information about smart rules, see [Smart rules](#).

DEVELOPING CUMULOCITY WEB APPLICATIONS

If you develop a Cumulocity web application using the [Web SDK](#), for an Edge appliance configured with a certificate not trusted by Node.js (for example, a self-signed certificate), then you must ensure that Node.js trusts the root certificate.

To add the Edge appliance's self-signed certificate to the Node.js trust store, set the environment variable `NODE_EXTRA_CA_CERTS` to the path of the certificate before executing the `npm` commands.

Example:

On Windows Powershell:

```
$env:NODE_EXTRA_CA_CERTS=<path-to-the-certificate>\certificate.pem
```

On Windows command prompt:

```
set NODE_EXTRA_CA_CERTS=<path-to-the-certificate>\certificate.pem
```

On Linux platforms:

```
export NODE_EXTRA_CA_CERTS=<path-to-the-certificate>/certificate.pem
```

For more information on the Cumulocity web application development, see [Web SDK](#).

WORKING WITH DATAHUB

ANALYTICAL QUERYING WITH DATAHUB EDGE

DataHub Edge is an optional component of Edge. DataHub Edge complements the ad-hoc querying of recent device data with analytical querying over long periods of time. For that purpose, data is moved from the Operational Store of Edge to a local data lake, with the data being stored in a concise and query-efficient format. DataHub Edge allows you to run SQL queries against the data lake contents so that you can gain more insights into your device data.

DataHub Edge is the counterpart of DataHub, the variant for cloud deployments. To learn more about DataHub in general, see [DataHub overview](#). To learn more about DataHub Edge, see [Running DataHub on the Edge](#).

 > Cumulocity Edge > Working with DataHub

DATA BROKER

Find out about the installation of the Messaging Service and the microservice-based data broker on Cumulocity Edge.

INSTALLING THE MESSAGING SERVICE AND THE MICROSERVICE-BASED DATA BROKER

The microservice-based data broker is an optional component of Edge. The data broker is a Cumulocity feature that allows a Cumulocity tenant to send data in motion (live measurements, events and alarms), using the standard Cumulocity REST API requests, to another tenant as that data arrives in the first tenant. It is often used to 'ship' data from remote Edge Cumulocity instances in the field to a central Cumulocity instance, where all data from all remote 'field' locations can be seen and handled together. Another use case is when devices are leased. The devices are managed by the leasing company in its tenant, and the devices' measurements and other data are sent to the leasing tenant via a data broker connection for application data processing.

The bundle for installing the Messaging Service and the microservice-based data broker comes as a `.tgz` file which contains the Messaging Service and data broker files, an install script and a digital signature. This bundle is unpacked, the signature verified and the install script executed by another script that is part of the standard Edge installation. The bundle can be downloaded from the [Cumulocity Download Center](#). The bundle is named "Cumulocity Data Broker Edge".

✓ REQUIREMENTS

Ensure that [microservice hosting feature is enabled](#) for Edge as both the Messaging Service and data broker run as Kubernetes pods. Running microservices on a basic Edge installation requires the number of CPU cores to be increased from 2 to 4 and the RAM to be increased from 6GB to 8GB. Running the Messaging Service and data broker requires an extra 4GB of RAM on top of this.

The Messaging Service microservice uses API version 1. To install the Messaging Service on Edge version 10.15, you must subscribe the "edge" tenant to the **Feature-privileged-microservice-hosting** application before uploading the Messaging Service microservice with API version 1. For more information, see [Deploying microservices with a lower manifest version](#).

Using SCP, copy the bundle to the `/tmp` directory on the Edge server. Enter the password for the Edge Linux administrative/login account when prompted. For example:

```
scp pulsar-edge-install-1018.0.0.tgz "<LOGIN_USERNAME>@<DNS_NAME_OF_SERVER>:/tmp"
```

Substitute the relevant values of `<LOGIN_USERNAME>` and `<DNS_NAME_OF_SERVER>`. Then run:

```
ssh <LOGIN_USERNAME>@<DNS_NAME_OF_SERVER> -t "/bin/bash -c 'cd /tmp && sudo /opt/c8y/utilities/install_signed_package.sh /tmp/pulsar-edge-install-1015.0.0.tgz'"
```

Enter the Edge Linux admin account password, again followed by the Cumulocity admin account password.

To confirm the Messaging Service and data broker are now running, log into the Edge VM using SSH with the connection details used previously and run:

```
sudo kubectl get pods -A
```

The command returns something like this:

NAMESPACE	NAME	READY	STATUS	RESTARTS	AGE	
c8y-messaging-service	pulsar-bookie-0	1/1	Running	0	4m14s	
c8y-messaging-service	pulsar-bookie-init-765k9	0/1	Completed	0	4m14s	
c8y-messaging-service	pulsar-broker-0	1/1	Running	0	4m14s	
c8y-messaging-service	pulsar-pulsar-init-z97xs	0/1	Completed	0	4m14s	
c8y-messaging-service	pulsar-zookeeper-0	1/1	Running	0	4m14s	
calico-system	calico-kube-controllers-5ddcc6fc8f-sln6x	1/1	Running	0	82m	
calico-system	calico-node-bxm9b	1/1	Running	0	82m	
calico-system	calico-typha-7fd9df674f-24zwz	1/1	Running	0	82m	
cumulocity-single-node	databroker-agent-server-scope-edge-deployment-7dc4dbdc5f-wxh8l	1/1	Running	3	3m36s	
cumulocity-single-node	device-simulator-scope-management-deployment-78d5c6694-rlpjx	1/1	Running	0	80m	
cumulocity-single-node	kube-registry-persistent-secure-7db46f57b5-vhlbh	1/1	Running	0	81m	
kube-system	coredns-74ff55c5b-5zkps	1/1	Running	0	82m	
kube-system	etcd-localhost	1/1	Running	0	82m	
kube-system	kube-apiserver-localhost	1/1	Running	0	82m	
kube-system	kube-controller-manager-localhost	1/1	Running	0	82m	
kube-system	kube-proxy-92lm6	1/1	Running	0	82m	
kube-system	kube-scheduler-localhost	1/1	Running	0	82m	
kube-system	metrics-server-5b78d5f9c6-7l45k	1/1	Running	0	82m	
tigera-operator	tigera-operator-76bbcbcb85-6clrf	1/1	Running	0	82m	

The key pods that should be running are `pulsar-bookie-0` , `pulsar-broker-0` , `pulsar-zookeeper-0` and `databroker-agent-server` , with a suffix for the pod name specific to the installation. It may take a few minutes for the Kubernetes pods to settle into a running state after the installation has been completed.

For more information on installing and troubleshooting the Messaging Service, see the Messaging Service installation & operations documentation] available on the [Cumulocity Download Center](#) (credentials required).

For more information on configuring and using the microservice-based data broker, see [Microservice-based data broker](#).

UNINSTALLING THE MESSAGING SERVICE AND THE MICROSERVICE-BASED DATA BROKER

The bundle for the Messaging Service and data broker microservice does not have an uninstall command. If a mistake is made or an installation fails, the bundle components must be uninstalled manually. Afterwards, the installation can be reattempted.

TO UNINSTALL MANUALLY

1. Login to Edge using SSH.
2. Uninstall the Pulsar Helm chart.

```
sudo helm uninstall pulsar -n c8y-messaging-service
```

3. Wait for the pulsar pods to terminate.
This command updates its state every two seconds. When it reports that there are no resources, press Ctrl+C to return to the command line.

```
watch sudo kubectl get pods -n c8y-messaging-service
```

4. Delete Pulsar's persistent volume claims.

```
sudo kubectl delete pvc -n c8y-messaging-service --all
```

5. Find the names of the persistent volumes used by the Pulsar components.

Here the grep command is used to filter the output. The names are in the output's leftmost column.

```

sudo kubectl get pv | grep pulsar
journal0 2Gi RWO Retain Released c8y-messaging-service/pulsar-bookie-journal-pulsar-bookie-0 local-storage 3h
ledgers0 10Gi RWO Retain Released c8y-messaging-service/pulsar-bookie-ledgers-pulsar-bookie-0 local-storage 3h
zookeeper0 2Gi RWO Retain Released c8y-messaging-service/pulsar-zookeeper-data-pulsar-zookeeper-0 local-storage 3h

```

6. Delete each of the persistent volumes found in the previous step.

```
sudo kubectl delete pv journal0 ledgers0 zookeeper0
```

7. Delete the directories used by the persistent volumes.

```
sudo rm -rf /opt/bookie/ledgers /opt/bookie/journal /opt/zookeeper
```

8. Unsubscribe the "edge" tenant from and delete, the databroker-agent-server microservice.

The databroker-agent-server application ID is required for the HTTP requests used to do both of these.

To find its application ID, you can use a cURL command of the following form and redirect the output to a jq command to make it easier to read the JSON response. The general structure of the command is:

```
curl http://<EDGE-HOSTNAME>/application/applicationsByName/databroker-agent-server -su <TENANT>/<USERNAME> | jq --indent 2
```

The ID required is the value of the `id` field at the root level of the JSON response. In the following example, it has the value '19' (the penultimate JSON field).

```

curl http://myown.iot.com/application/applicationsByName/databroker-agent-server -su edge/admin | jq --indent 2
Enter host password for user 'edge/admin':
{
  "next": "https://myown.iot.com/application/applications?pageSize=5&currentPage=2",
  "self": "https://myown.iot.com/application/applications?pageSize=5&currentPage=1",
  "statistics": {
    "pageSize": 5,
    "currentPage": 1
  },
  "applications": [
    {
      "owner": { ... truncated ... },
      "requiredRoles": [ ... truncated ... ],
      "manifest": { ... truncated ... },
      "roles": [],
      "contextPath": "databroker-agent-server",
      "availability": "MARKET",
      "type": "MICROSERVICE",
      "activeVersionId": "1345",
      "name": "databroker-agent-server",
      "self": "https://myown.iot.com/application/applications/19",
      "id": "19",
      "key": "databroker-agent-server-key"
    }
  ]
}

```

Use that ID value in the request to unsubscribe from the microservice and again when deleting it.

To unsubscribe, use a command of the following form:

```
curl -X DELETE http://<EDGE-HOSTNAME>/tenant/tenants/<TENANT>/applications/<APP-ID> -u <TENANT>/<USERNAME> -v
```

This returns an HTTP 204 (no content) response. For example:

```

curl -X DELETE http://myown.iot.com/tenant/tenants/edge/applications/19 -u edge/admin -v
Enter host password for user 'edge/admin':
* About to connect() to myown.iot.com port 80 (#0)
* Trying 127.0.0.1...
* Connected to myown.iot.com (127.0.0.1) port 80 (#0)
* Server auth using Basic with user 'edge/admin'
> DELETE /tenant/tenants/edge/applications/19 HTTP/1.1
> Authorization: Basic ZWRnZS9hZG1pbjpfZGdlN184OV9fX19fX18=
> User-Agent: curl/7.29.0
> Host: myown.iot.com
> Accept: */*
>
< HTTP/1.1 204 No Content
< Date: Thu, 07 Sep 2023 16:27:32 GMT
< Connection: keep-alive
< X-Content-Type-Options: nosniff
< X-XSS-Protection: 1; mode=block
< Cache-Control: no-cache, no-store, max-age=0, must-revalidate
< Pragma: no-cache
< Expires: 0
< X-Frame-Options: DENY
<
* Connection #0 to host myown.iot.com left intact

```

To delete, use a command of the following form:

```
curl -X DELETE http://<EDGE-HOSTNAME>/application/applications/<APP-ID> -su <TENANT>/<USERNAME> -v
```

This returns an HTTP 204 (no content) response. For example:

```

curl -X DELETE http://myown.iot.com/application/applications/19 -su edge/admin -v
Enter host password for user 'edge/admin':
* About to connect() to myown.iot.com port 80 (#0)
* Trying 127.0.0.1...
* Connected to myown.iot.com (127.0.0.1) port 80 (#0)
* Server auth using Basic with user 'edge/admin'
> DELETE /application/applications/19 HTTP/1.1
> Authorization: Basic ZWRnZS9hZG1pbjpfZGdlN184OV9fX19fX18=
> User-Agent: curl/7.29.0
> Host: myown.iot.com
> Accept: */*
>
< HTTP/1.1 204 No Content
< Date: Thu, 07 Sep 2023 16:35:11 GMT
< Content-Type: application/vnd.com.nsn.cumulocity.error+json; charset=UTF-8; ver=0.9
< Connection: keep-alive
< X-Content-Type-Options: nosniff
< X-XSS-Protection: 1; mode=block
< Cache-Control: no-cache, no-store, max-age=0, must-revalidate
< Pragma: no-cache
< Expires: 0
< X-Frame-Options: DENY
<
* Connection #0 to host myown.iot.com left intact

```

REST APIS FOR EDGE

For information about the REST APIs for Edge, see the [Cumulocity Edge OpenAPI Specification](#).

! IMPORTANT

Wherever possible, the REST resources and its representations are maintained in a *backwards compatible* manner. However, the Edge installation or configuration scripts developed based on the REST API of an older Edge version (for example, 10.9.0.0) may not work against the newer version (for example, 10.15.0.0). Also, the behavior of the REST API may change without prior notice if the existing behavior is incorrect or constitutes a security vulnerability.

This API is not covered by the general Cumulocity API backwards compatibility statement mentioned in [Compatibility policy](#).

INTRODUCTION

Learn about the conceptual aspects of Cumulocity Edge, the single-server variant of the Cumulocity platform.

OVERVIEW

Edge is delivered as a software appliance designed to run on industrial PC's or local servers.

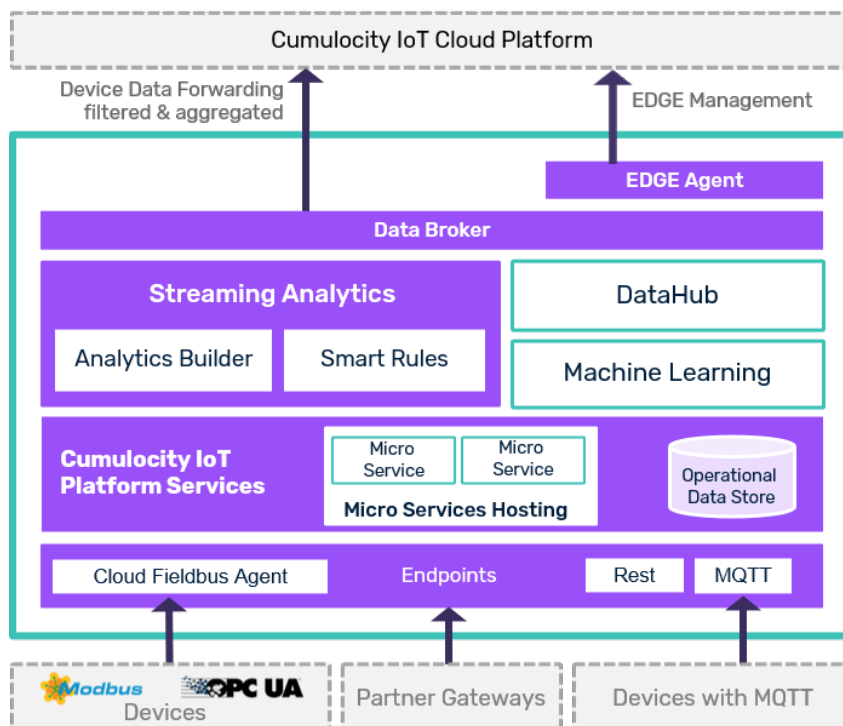
In contrast to Cumulocity platform, which is available in the cloud (for example, using AWS, Azure or other data centers), Edge is installed in factories, that is, in the same site ("onsite") in which the IoT assets are located.

Reasons for using an onsite installation of Edge include:

- **Autonomy:** Even if there is no cloud connection, tasks like data collection and data analysis can still be performed.
- **Data reduction:** Data is analyzed and aggregated close to assets, and thus less data needs to be sent to the cloud.
- **Reactivity:** Both Edge and Cumulocity platform include real-time streaming analytics engines. However, placing the rule execution in Edge reduces latency, because the round-trip to cloud is omitted.

Features of Edge include:

- Edge Agent, which enables remote monitoring and management of an Edge instance from the Cumulocity tenant.
- Data Broker to send IoT data to the cloud and receive operations from the cloud, with web-based UI to filter data.
- Streaming Analytics engine for real-time local data analysis including the Cumulocity Analytics Builder.
- Ready-to-use Cockpit and Device Management applications.
- Native protocol support for MQTT and REST.
- Edge database for operational data storage.
- **OPC UA** with web-based UI for efficient connection management and seamless integration of industrial automation systems.
- Easy installation, upgrades and backup/restore.
- Microservice hosting, which allows to run server-side applications which may be used to extend the Cumulocity platform with customer-specific functionality (optional component).
- **DataHub Edge** for historical data storage and analytics (optional component).



VERSIONS

Cumulocity Edge 10.18 uses the following versions:

Edge component	Version
Cumulocity platform	10.18.0
Streaming Analytics	10.18

CUMULOCITY EDGE VERSUS OTHER CUMULOCITY DEPLOYMENTS

Edge uses the same software as Cumulocity platform.

However, while the base software is the same, there are differences regarding the activated optional features and pre-installed agents.

The following differences apply:

Area	Edge	Cumulocity platform
Multi-tenancy	No; single tenant	Yes
Cluster	No; single server	Yes
High availability	HA capabilities depend on the underlying virtualization technology, server failure could lead to temporary downtime*	Full HA: No downtime on server failure, optionally even for data center failure
Vertical scalability	Yes, limited to appr. 100 tps per CPU core	Yes, but not used
Horizontal scalability	No	Yes, nearly unlimited scalability
Upgrades with no downtime	No	Yes
Root access	Yes	Yes, if customer is hosting
Installation	Offline, with VM image	Online
Cloud Field Bus	Included	Optional
Streaming Analytics	Included	Optional
Data Broker	Included	Optional
OPC UA	Included	Optional
Microservice Hosting	Optional	Optional

Area	Edge	Cumulocity platform
Machine Learning	Optional	Optional
Data Hub	Optional	Optional

Footnote: * The underlying infrastructure including the virtualization is not part of the product. For further details on high availability or fault tolerance options, refer to the relevant information provided by third parties (for example, [vSphere Availability](#)).

In addition to Edge, we provide a light-weight approach to implement Edge functionality: the open-source Cumulocity Device SDKs. These can be utilized to build agents on less powerful devices like embedded devices.

EDGE FUNCTIONALITIES

Since Edge is based on the same software as the cloud-based Cumulocity platform version, the included applications are the same in both versions, with minor restrictions.

For general information about functionalities and features of Cumulocity platform, see [Get familiar with the UI](#).

Below you can find the links to the most common functionalities of Cumulocity.

DEVICE MANAGEMENT

The [Device Management application](#) provides functionalities for managing and monitoring devices and enables you to control and troubleshoot devices remotely.

The following functionalities are available in Edge and described in detail in these sections:

SECTION	CONTENT
Registering devices	How to register one or more devices manually and how to bulk-register devices in order to connect devices to your account.
Viewing all devices	What is displayed in the device list and how to sort devices by filtering devices .
Grouping devices	Why and how to group devices into top-level groups, subgroups and smart groups .
Viewing device details	Detailed description of the various kind of information available for various types of devices.
Monitoring and controlling devices	How to monitor the connection quality and availability status of devices, how to handle alarms from devices, how to remote control and how to troubleshoot devices.
Managing device services	How to monitor the data that your devices send about the services they are running. See detailed information about measurements , events and alarms sent for every service.
Managing device types	How to process data from various device types by using device protocols.

SECTION	CONTENT
Managing device data	How to retrieve and manage firmware and software for devices; how to retrieve configuration data and store and manage it in a configuration repository as configuration snapshots.
SmartREST T templates	How to work with SmartREST templates , a collection of request and response templates used to convert CSV data and Cumulocity Rest API calls.
Working with simulators	Create a simulator with the same level of functionality as connected hardware devices and simulate messages (measurements, alarms, events and inventory) that the device sends to the Cumulocity platform with series of instructions that the simulator executes one after the other.
Connectivity	How to set up your Jasper Control Center account , configure the connectivity for the SIM provider in your Cumulocity tenant, link SIMs and mobile devices , and checking connectivity .

COCKPIT

The [Cockpit application](#) provides you with options to manage and monitor Internet of Things (IoT) assets and data from a business perspective.

The following functionalities are available in Edge and described in detail in these sections:

Section	Content
Managing assets	Organize assets in hierarchies by creating groups and assigning devices .
Data Explorer	Interactively explore, compare and visualize IoT data. Describes how to access and use the data explorer , add data points to the data explorer, customize data point properties , modify the visualization , store the data explorer as widget , and export the data.
Working with dashboards	Create your own analytics and monitor pages by adding and arranging widgets . Share dashboards among all devices of the same type.
Working with reports	Handle reports based on dashboard layouts, create reports for exporting data in CSV or excel format and schedule the export .
Using widgets in dashboards and reports	Widgets can display maps, images, graphs, tables and other graphic representations of data. Add a widget to a dashboard or a report , modify widgets, and add assets to a new or existing widget .
Widgets collection	Use various types of widgets from the Widgets collection that comes with Edge and configure them according your needs.
Managing exports	Export data to either CSV or Excel files. Add and schedule an export to a CSV or XLSX file to any point in time.
Data point library	Manage default settings ("profiles") of your devices and apply them automatically using the Data point library .
Smart Rules	Create and manage business rules to work on incoming data in realtime and to perform actions based on this data.
Smart Rules collection	Use pre-defined global Smart Rules to configure rules for geofencing, thresholds or alarm escalation and notifications (SMS/email). Describes each smart rule and its configurable parameters in detail.
Configuring Cockpit applications	Configure custom Cockpit applications according to your individual needs.

ADMINISTRATION

The [Administration application](#) enables account administrators to manage their users, roles, tenants and applications and lets them configure a number of settings for their account.

The following functionalities are available in Edge and described in detail in these sections:

SECTION	CONTENT
Configuring Edge	Providing information on Edge specific configuration options.
Home screen	Providing information on your capacity usage and subscribed applications.
Managing users	How to add users , edit, disable or delete them.
Managing permissions	How to create and edit global roles and inventory roles , how to assign them to users, and how to grant application access .
Managing applications	How to manage and configure own applications in your Edge account.
Audit logs	How to view and filter audit logs that show security-relevant operations that have been processed by a user. See the different audit log types for more information.
Alarm mapping	Change the severity and text of alarms to adapt them to your business priorities.
Managing data	How to manage and configure retention rules for your data and how to manage stored files in the file repository.
Changing settings	How to change account settings like application settings , manage the properties library , enable SMS provider features like user notifications and two-factor authentication, and manage credentials for different providers .